Currency Returns and Price Discovery around Global FX Fixings

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ABSTRACT

The research looks at the possibility of a W-shaped return pattern, in which the dollar climbs before FX fixes and falls after them, in an effort to shed light on the daily movements of the U.S. dollar during FX fixing periods. This is accomplished by analyzing the hourly returns of the nine most traded currencies over a span of 21 years. To determine the reason for the shift in return orientation, they use statistical analysis and natural testing. Based on the daily fluctuations in spot trading volume of over one billion USD, we can see that return reversals occur often and statistically significantly for all major currencies. The availability of published reference rates largely dictates the timing of these changes. The research also shows that the risk that foreign exchange merchants take on with their stock is the root cause of these price fluctuations. This is important for regulators, policymakers, and central banks to remember so they can make foreign currency markets more stable and effective and better understand how exchange rates fluctuate over time.

Keywords: FX Fixings; Price Discovery; Currency Returns; Return Reversals; Spot Trading Volume.

Introduction

Globally, no financial market is as large or as adaptable as the foreign currency (FX) market. All hours of the day and night, it facilitates investments, international trade, and capital transfers. Although the forex market is notoriously accommodating, rates may often fluctuate throughout the day in ways that defy logic. The behavior of exchange rates during the times when benchmark currencies are fixed is one example of these peculiarities. Institutional investors, MNCs, asset managers, and central banks frequently utilize them during these periods for portfolio evaluation, trade execution, and financial reporting. During these moments, a lot of activity occurs at once, which may lead to huge order imbalances that can momentarily screw with exchange rates. Fixing benchmarks, on the other hand, should make reference pricing very obvious and constant. During periods of heavy trading, these distortions might lead to problems with inventory risk, market

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efficiency, liquidity, and currency pricing, as well as make market trading more costly for individuals. Research into market microstructure and global finance cannot proceed without an understanding of the impact of benchmark fixes on day-to-day fluctuations in exchange rates. Intraday trading in foreign exchange markets has been the subject of much study, with a focus on regional trading sessions, fresh information arrivals, volatility, trading volume, and bid-ask spreads. Opening and closing times for the main financial markets have fluctuated daily. Newer research, however, has zeroed in on how trade activity and the volatility of currency rates across time zones relate to one another. More recent research found greater volatility, abnormal trading volumes, and short-term price pressure in benchmark fixing windows, particularly the London fix. Additionally, research has shown that standard rates may influence short-term fluctuations in exchange rates and that demand for US dollars remains constant at certain set intervals. In contrast, the majority of existing research has focused on either individual fixing events or tiny event dates or markets. These restrictions prevent us from gaining a comprehensive understanding of the role that tandem benchmark fixes have in the development of hourly prices throughout the trading day. There is some evidence that benchmark modifications and short-term fluctuations in exchange rates are connected, but there are still some key flaws in the study. To start, prior research that concentrated on the London fixing window neglected the Tokyo and ECB fixes, as well as their cumulative effects on the global currency markets. That implies there hasn't been enough real-world study on how numerous test fixes occurring at the same time in various time zones affect each other. Second, most studies look at pairings of currencies and not a trading Strategy's that examines how the U.S. dollar changes frequently versus a big group of other currencies. Because of this myopic view, it is more difficult to determine if the seen patterns of returns are indicators of a larger systemic aspect of the global foreign exchange markets or whether they are just isolated occurrences affecting a single currency. Even when there are common events like regime transitions, DST shifts, and market breakdowns, the duration of these return reversals remains a mystery to us. (Ahmad et al., 2024; Alper et al., 2024)

Results that are tied to benchmarks might move in the other direction due to the way the economy operates, which is another significant constraint. Knowledgeable traders can potentially get an advantage before the benchmark shift based on private knowledge, according to traditional ideas of asymmetric information. The fact that the announcement of planned policies and macroeconomic news does not always coincide with the fluctuations in the value of the dollar

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during window dressing means that there is little empirical evidence to support this theory. Data from high-frequency transactions are not strongly associated with dealer inventory risk, order imbalances, liquidity provision, or conditional return reversals in studies. A possible substitute, however, is provided by theories of inventory management, which center on the compensation that dealers get for temporarily bearing order imbalances. It is difficult to determine whether price increases are temporary educational benefits or compensation for inventory risk since previous research often fails to distinguish between those who are supplying liquidity and those who are seeking it. The mechanics of regular reversals in exchange rates tied to benchmarks are still a mystery to us. (Bakshi et al., 2024; Asimakopoulos et al., 2024)

Using high-frequency data for G9 currencies, this research contributes to the existing literature by examining the daily fluctuations in exchange rates around the three major standard currency fixes in Tokyo, Frankfurt (ECB), and London. This data strongly support the claim that systematic reversals in US dollar returns occur and remain in place throughout the trading day globally. The article also divides liquidity supply and demand returns and examines the role that dealer inventory management and order imbalances may play in explaining these trends. The research sheds light on the economic variables that influence changes in benchmark-related exchange rates and how daily FX prices are created by combining benchmark-setting behavior with market microstructure theory. (Aretz et al., 2025; Baele et al., 2025)

I. FX Benchmarks

The foreign exchange market is "fixed" at a certain moment each day when the sum of all bids and offers is announced. Historically, the three most popular solutions have been those in Tokyo, London, and the ECB. Figure 1, labeled "around the clock" in ET (New York time), clearly displays the changes. Figure 1's coloured blocks indicate the times of day when each region's futures market is open for trading. The number begins at five o'clock in the afternoon Eastern Time, which is about when the trading day begins in Australia and New York City. At 9:55 A.M. local time (or 20:55 ET when daylight saving time is not in force in the US), the first major currency fix takes place in Tokyo. The overlap between the red, green, and yellow blocks represents the fact that trade in Japan concludes at the end of the day. Even if the ECB fix is at 8.15 ET (14.15 local time), which is near the start of the trading day in New York (let's say 8.00 ET), the timing is plainly off. Importantly, the ECB shift does not occur at 8:30 ET, the hour

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typically reserved for macro remarks. Below, we'll demonstrate how this time gap is significant in understanding the day-to-day fluctuations in exchange rates. London will host the day's last and most crucial fix at 16:00 local time (11:00 ET).

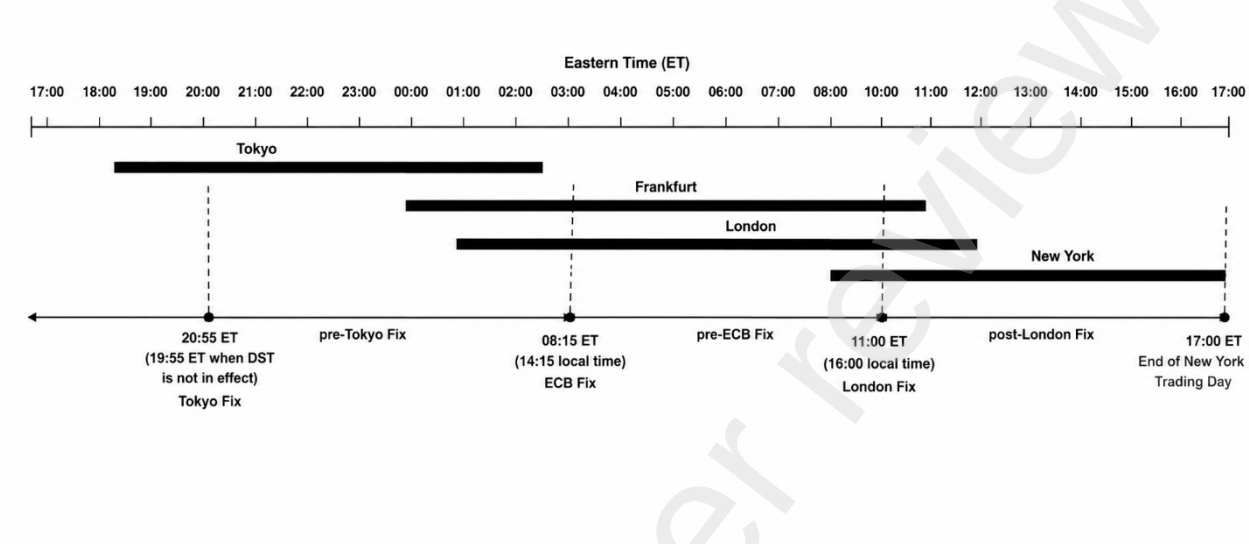


Figure 1: Time zone corrections for currency. [You may see the colour figure at

Although there is some variation in the methods used to establish them, the timing of their release, and the organizations that exercise control over fixes, all of them affect the foreign currency markets.

Every bank has its own predetermined rate that it charges its clients. The Tokyo fix rates are announced at 10:00 local time. When contrasted with the ECB and London fixes, where a single reference rate is distributed, the key distinction here is substantial. Rates for the Tokyo fix are determined by banks using their own client transactions at 9:55 A.M. Additionally, the set pricing applies to any sales made after 10:00 from both prefix and postfix customers. The remainder of the trading day is dominated by the Tokyo fix rate, which banks closely monitor. Secondly, every day at 14:15 CET, the heads of the central banks in the Eurozone come together for a call where they discuss potential reference rates. This is how the ECB gets its start. By averaging the euro's bid and offer prices, the European Central Bank's reference rate is determined. We don't utilize actual trades to get the reference rate. However, the ECB reference rates are often used by non-financial enterprises in the Euro region that employ forward contracts for risk management. The European Central Bank (ECB) changed the timing of reference rate releases from 10:00 CET to 16:00 CET in July 2016, but otherwise kept the same procedure. They did this

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to discourage traders from relying on the ECB fix and to emphasize that the Euro FX reference rates are intended only for informational reasons. Because businesses relied on the ECB fix for cross-border contract valuation, protection, and settlement, Thomson Reuters established the WM/Reuters 14:00 CET standard. (Batten et al., 2024; Hattori & Shimizu, 2024)

As a last step, at 16:00 London time, Thomson Reuters announces the WM/Reuters London fix rate. For the purpose of calculating average rates, transactions, including quotations for less liquid currency pairings, are analyzed within a time frame that begins at around 16:00. From 15:57:30 to 16:02:30, all of the trades that occurred within five minutes of the fix may be located by using the bid and offer rates to discover the middle trade. The largest FX platforms have their trading rates retrieved every second. Furthermore, unlike the Tokyo Fix, the London Fix is reserved for orders submitted before 16:00 local time and intended to be executed at the benchmark rate. When valuing their overseas stock holdings, many market participants turn to the London Fix. (Melvin & Prins, 2024; Ranaldo & Santucci de Magistris, 2025)

II. Data

Data originates from various sources, including the Chicago Mercantile Exchange (CME), the Intercontinental Exchange (ICE), Refinitiv Tick History (RTH), and Olsen Financial Technologies. Here we provide a concise synopsis of the key points. The Internet Appendix I provides more details on data preparation, data cleansing, and data source discovery. From January 1999 through December 2019, the whole sample contains high-frequency tick-by-tick data on the G9 currencies. The following currencies are in circulation: the Australian dollar (AUD), the Canadian dollar (CAD), the euro (EUR), the yen (JPY), the dollar of New Zealand (NZD), the krone (NOK), the krona (SEK), the franc (CHF), and the pound (GBP). According to the latest BIS Triennial Survey, these currencies accounted for almost 75% of all daily foreign exchange trading and were the most actively traded during the research period. (Boffelli & Urga, 2024; Ito & Hashimoto, 2024)

From January 1999 through December 2019, RTH provided us with several indicative bid and ask prices. To facilitate "bank-to-client" trading, RTH compiles quotations from several financial institutions. Even though the majority of dealings in the foreign exchange market still take place over the counter, the RTH data does not reveal the frequency or value of these transactions. We have also had the ability to trade quantities and prices in real time from the RM site since June

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2006. We can learn more from the RM data and apply it to other order flow metrics as well. Major foreign exchange (FX) interdealer systems include Electronic Broking Services (EBS) and RM. In contrast to EBS's daily trading volume of 76 billion USD, RM's volume exceeds 100 billion USD. While not all currency pairs exhibit equal liquidity on both platforms, their results show a high correlation. On both platforms, their results are highly correlated. If we use Commonwealth currencies as an example, the most liquid platform would be RM. (Hegazy & Moore, 2025)

III. Currency Returns around the Clock

This section offers fresh data on the intraday fluctuation of currency returns. We establish a new stylised fact: the US dollar increases before FX fixes and then declines, according to a predicted pattern of intraday returns on exchange rates (Lyócsa et al., 2024; Gargano et al., 2024)

A. Currency Data and Sample Description

Denote by s_t a number that shows how much one US dollar is worth in other currencies, expressed as a logarithm, and Δs_t the change in the log exchange rate between time $t-1$ and t. A negative Δs_t shows how much the value of the US dollar has grown compared to other money. Daily close-to-close log spot results are given in Eastern Time (ET). $\Delta s_t^{\mathrm{CTC}}$ as the percent change in the mid-quote from 17:00 on day $t-1$ to 17:00 on day t,

$$\Delta s_t^{\mathrm{CTC}} = s_t^{17:00} - s_{t-1}^{17:00}. \quad (1)$$

The day was then divided into three halves according to the three main global currency fixes: the ECB at 14:15 a.m., London at 16:00 a.m., and Tokyo at 9:55 a.m. All timings are in Eastern Time. Here is how we calculate returns for the five intraday periods: (i) the time before Tokyo (18:30–20:55); (ii) the time after Tokyo (20:55–2:00); (iii) the time before the ECB (2:00–8:15); (iv) From 8:15 to 11:00, the ECB fix is followed by the London fix; and (v) From 11:00 to 17:00, the post-London fix follows. Sixth, to differentiate between the periods after Tokyo and before the ECB fix, we use 8:00 Frankfurt time (2:00 ET), the start of the European business day. Also, at 8:00 ET, the New York Forex trading day begins.

B. Intraday Dynamics of Currency Returns

Starting from January 1999 through December 2019, we plot the average daily five-minute log returns from 17:00 ET to 17:00 ET. Holding EUR, GBP, and JPY yields an average cumulative

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169 return (in basis points), as shown in Figure 2. Figure 3 shows the HOURLY and cumulative returns
170 of the DOL (unconditional dollar portfolio), which maintains a constant value for all foreign
171 currencies.

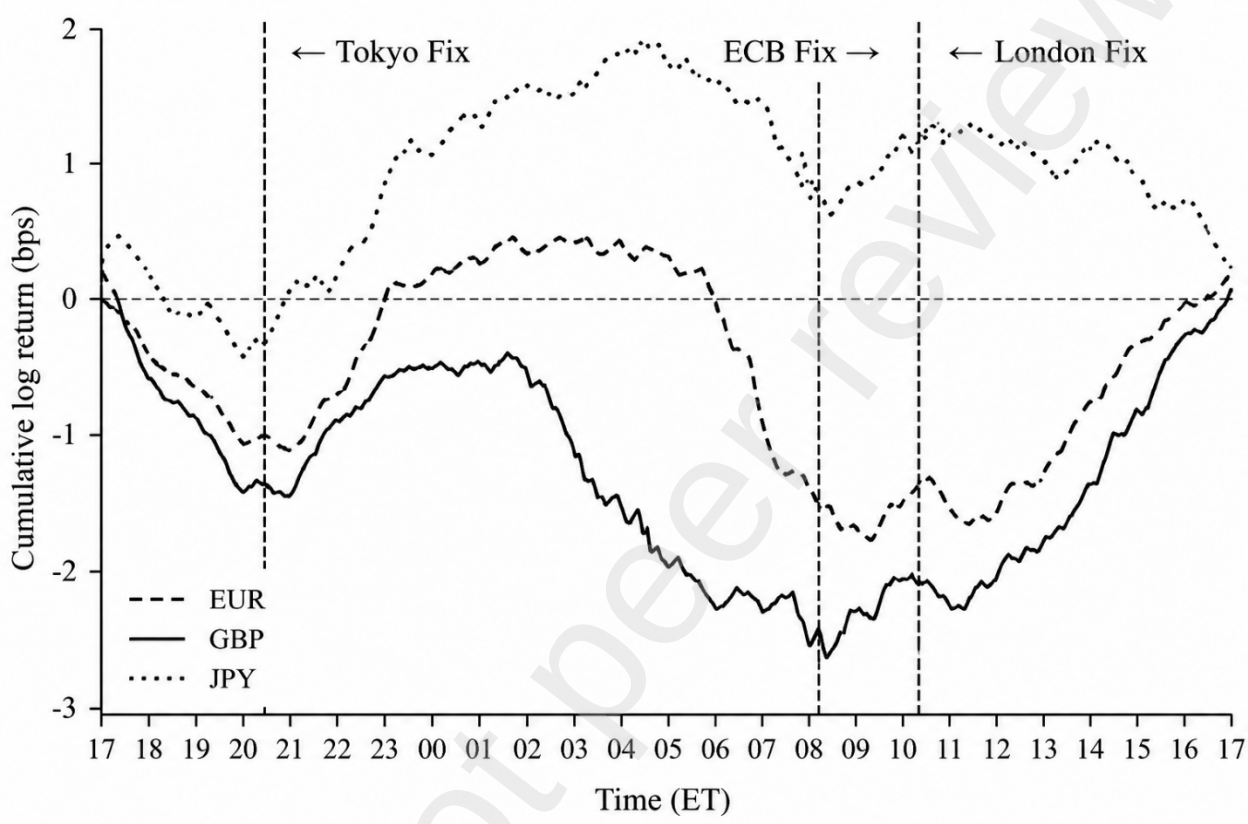


Figure 2: Returns on the EUR, GBP, and JPY over five minutes. This is a graph of the average five-minute results over time. Δs The chart also includes the EUR (dashed line), GBP (dash-dotted line), and JPY (dotted line) for a single working day. The value of the other currency relative to the US dollar increases as its value rises. At 20:55, 8:15, and 11:00, the three black dashed lines represent the prices in Tokyo, the ECB, and London, respectively. When discussing returns, one uses basis points. To tell the time, one uses ET. January 1999 through December 2019 is the time span covered by the sample. Finality Tick Records (Refer to the colour chart at

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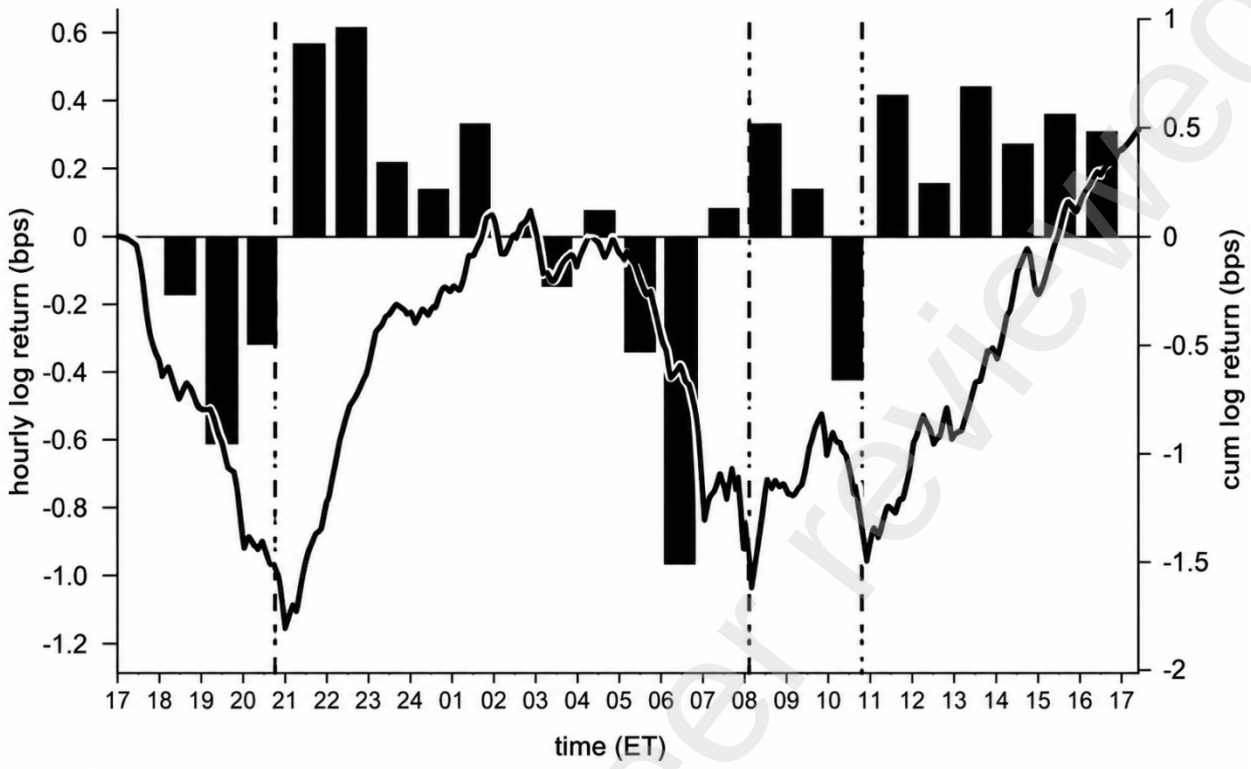


Figure 3: Variations in Intraday Dollar Return on Portfolio This figure represents the average amount of five-minute returns within a certain time period. Δs The goal is to engage in daytime trading using the dollar account. If it rises, it indicates that the group of currencies is becoming more powerful in relation to the US dollar. The Tokyo, ECB, and London fixings Three thin black lines at 20:55, 8:15, and 11:00 represent the Tokyo, ECB, and London fixings, respectively. by three thin black lines at 20:55, 8:15, and 11:00, respectively. When discussing returns, one uses basis points. To tell the time, one uses ET. From January 1999 to December 2019, this sample covers a wide time span. Results from the Refinitiv Tick historical database.

When compared to the US dollar, all currencies are clearly falling until the 20:55 ET Tokyo fix. Then they will all rise. Prices are falling relative to the US dollar as we approach the ECB fix. The European market opens at 2:00 ET, and the euros are falling more rapidly than the dollars of Australia, New Zealand, and Canada. There doesn't seem to be a trend in the cross section between the ECB fix and the London fix, except for EUR and JPY, which go up until one hour before the London fix. All currencies have changed a lot in relation to the US dollar since the London fix. So long as the USD doesn't go in the other direction of the JPY, the trend should last until the US

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work day ends at 17:00 ET. During the U.S. trading day, most currencies gain value while the Japanese yen loses it. 7

Since the value of foreign currencies tends to decline before the Tokyo fix and after the opening of the European markets, as well as the US dollar's worth tending to decline during the day, the total return for the whole day follows a typical W shape. (Baruník et al., 2024; Tiwari et al., 2024)

Every two trading days, the value of the dollar fluctuates by about 2 basis points. When compared to the currency spot market, the total return is on the big side. We discover that daily variations of more than $1 trillion are typical based on the daily turnover numbers from the 2019 BIS Triennial Survey. Table I presents the average FX log returns (changes in exchange rates) and t-statistics for the several daily subperiods we covered earlier, which helps to put the results in Figures 2 and 3 into a more formal context. 8

Table I. Intraday Returns and the Tokyo, ECB, and London Fixes

Throughout the day, this table displays the average values in basis points (bps) around the Tokyo, ECB, and London prices. **T-tests** are included in the supplementary data. Each time period is defined in Appendix Note 1. From January 1999 to December 2019, this sample covers a wide time span. Results from the Refinitiv Tick historical database.

Variable	Pre-T	Post-T	Pre-E	E-L	Post-L	CTC
AUD	−2.84	1.85	−0.41	−0.05	1.93	0.51
	(−8.24)	(3.94)	(−0.78)	(−0.11)	(3.75)	(0.48)
CAD	−1.52	1.50	−0.77	−0.41	1.55	0.38
	(−8.75)	(7.14)	(−1.94)	(−0.89)	(3.81)	(0.49)
CHF	−1.24	1.25	−2.55	1.27	2.24	1.02
	(−5.57)	(4.93)	(−4.19)	(2.55)	(5.19)	(1.05)
EUR	−1.80	2.34	−3.53	0.63	2.73	0.41
	(−8.84)	(9.29)	(−7.25)	(1.36)	(6.78)	(0.47)
GBP	−1.89	1.27	−2.31	0.42	2.64	0.18
	(−9.12)	(5.03)	(−4.65)	(1.03)	(7.43)	(0.22)
JPY	−1.61	3.15	−1.04	0.62	−1.16	−0.07
	(−5.88)	(8.22)	(−2.37)	(1.43)	(−3.03)	(−0.06)

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Variable	Pre-T	Post-T	Pre-E	E-L	Post-L	CTC
NOK	−1.94	2.95	−1.40	−2.01	2.14	−0.31
	(−7.52)	(9.67)	(−2.20)	(−3.76)	(4.48)	(−0.29)
NZD	−3.39	2.29	−0.35	0.56	1.81	0.97
	(−8.42)	(4.00)	(−0.61)	(1.10)	(3.30)	(0.89)
SEK	−1.80	2.16	−3.08	−0.05	2.60	−0.20
	(−6.53)	(7.06)	(−4.95)	(−0.09)	(5.38)	(−0.18)
DOL	−1.99	2.10	−1.68	0.14	1.84	0.45
	(−11.69)	(9.19)	(−4.51)	(0.38)	(5.48)	(0.63)

Prior to the Tokyo fix and after trading in New York finishes, the value of all other currencies vs the US dollar decreases. The value loss for the CHF/CAD combination is the lowest compared to other currency pairings. But with average returns of -2.8 and 3.4 basis points, respectively, the Australian dollar and the New Zealand dollar are the worst. When you set the significance level to 1%, you will see average returns greater than zero on all currency pairs. All of the currencies in our sample exhibit changes that are statistically significant after the Tokyo adjustment. At 3.0 and 3.2 basis points daily, respectively, the JPY and NOK see the most significant fluctuations. The DOL's strong and obvious inclination to reverse is likewise unsurprising. It experiences a daily loss of around 2 basis points until the Tokyo fix, after which it regains its losses.

A tremendous deal of value The yen and other European currencies lost significant value relative to the dollar before the ECB fix. The yen and other European currencies lost significant value relative to the dollar before the ECB fix. According to the data, the point predictions are very reliable. The EUR and SEK experienced a challenging day, with losses of −3.5 and −3.1 basis points, respectively. In the past, there was less consistency in the movement of currencies between the ECB and London fixes. Possible explanation: the time period encompasses both the ECB fix and the London fix, when the value of the dollar drops after the former and rises before the latter. This phenomenon is illustrated in Figures 2 and 3.

After the London adjustment the pattern becomes even more apparent: all currencies gain ground between the adjustment and the conclusion of the US trading day, with the exception of the JPY, which loses 1.2 basis points. The CAD and EUR gain 1.5 and 2.7 basis points, respectively.

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Overall, the DOL leads by 1.8 basis points, and all currencies exhibit significant statistical changes.

Our discovered pattern is consistent with daily seasonality, as seen in the final column of Table I; hence, it has a predicted component that is irrelevant to outcomes that are very near to the close. During the course of the research period, not a single currency in our group saw daily fluctuations above one basis point, and none of the returns that were near the close were statistically significant. (Ben-David et al., 2025)

C. Robustness Checks

We use two metrics to assess the stability of the reversals around the fixes: (i) throughout time and (ii) across datasets.

The intraday DOL returns are divided into four subsamples for the appropriate time periods in Table II. As we track the returns for each subsample across a 24-hour trading day, we see a W-shaped pattern. There is a large DOL reverse in every subsample. Between 1999 and 2014, the average daily change on either side of the fix was about 2 basis points. The reversal around the Tokyo benchmark from 2014 to 2019 is statistically significant but less than before. Returns on pre- and postfix pairings remain high over time and follow a pattern that averages down to zero, indicating that intraday FX seasonalities are a solid component of daily data. Daily changes are still rather little, however, so I'm starting to wonder whether the tendency is due to the fact that RTH-suggesting quotations are used to estimate log spot changes.

Table II. Intraday Dollar Portfolio Returns, Year-over-Year

The following table displays the average basis point returns for the dollar portfolio around the Tokyo, ECB, and London prices for daily periods. Every one of the sample periods is uniformly spaced. Refer to Note 1 in the appendix for a clarification on the meaning of the intraday periods. In square brackets, you can see the t-statistics. The study duration runs from January 1999 to December 2019. Results from the Refinitiv Tick historical database.

Period	Pre-T	Post-T	Pre-E	E-L	Post-L	CTC
1999–2003	−1.64	2.48	−3.43	1.32	2.80	1.61
	(−5.15)	(6.63)	(−4.84)	(1.98)	(4.57)	(1.25)

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2004–2009	−2.67	2.91	−1.50	−0.11	2.29	0.97
	(−6.92)	(5.88)	(−1.80)	(−0.13)	(3.09)	(0.63)
2010–2013	−3.49	2.31	−0.35	0.11	1.43	0.05
	(−8.38)	(3.87)	(−0.40)	(0.14)	(1.80)	(0.04)
2014–2019	−0.63	0.82	−1.25	−0.60	0.86	−0.84
	(−2.64)	(2.25)	(−2.28)	(−1.02)	(1.62)	(−0.80)

We utilize three distinct data sets—intraday returns for the DOL derived from RM value-weighted average prices (VWAPs), RTH forwards, and CME futures midquotes—to answer this question. To keep things simple, we'll just be looking at data from June 2006 through December 2019. Not only do the CAD, CHF, and SEK factors in, but so do the EUR, GBP, and JPY. 9 Our intraday findings for the ICE U.S. dollar index (USDX), a widely traded currency index, are similarly based on this same currency composition.

Since the amplitude of the reversals around the fixes determined from futures and spot prices is very similar, there is no intraday trend in implied interest rate differentials (Table III). The results from the CME and ICE futures markets are likewise quite significant statistically, and they agree with the main points in Table I; therefore, it's safe to say that the same holds true for the firm quotations from the electronic FX derivatives markets. While trends are visible in traded prices sourced from RM, they The effective bid-ask spread fails to capture the visible trends in traded prices sourced from RM. by the effective bid-ask spread. (Du et al., 2025; Chen et al., 2025)

Table III. Intraday Returns across Different Data Sets

The table shows the average basis point (bps) returns for the dollar portfolio around the Tokyo, ECB, and London fixings for various intraday periods. The meaning of intraday periods is defined in note 1 of the appendix. In parentheses, you can see the t-statistics. June 2006–December 2019 is the duration of the sample period. Sources of data include the following: spot and forwards prices from Refinitiv Tick History (RTH); futures from CME and ICE dollar index; and volume-weighted average prices (RM VWAPs) from Refinitiv FX Matching.

Variable	Pre-T	Post-T	Pre-E	E-L	Post-L	CTC
RTH Spot	−1.58	1.88	−1.65	−0.04	1.20	−0.21

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Variable	Pre-T	Post-T	Pre-E	E-L	Post-L	CTC
	(−8.16)	(7.32)	(−3.73)	(−0.10)	(3.08)	(−0.26)
RTH Forwards	−1.57	1.85	−1.61	−0.05	1.20	−0.20
	(−8.12)	(7.18)	(−3.65)	(−0.12)	(3.08)	(−0.25)
CME Futures	−1.46	1.80	−1.64	0.11	0.64	−0.59
	(−8.94)	(9.37)	(−4.76)	(0.30)	(2.20)	(−0.95)
ICE Futures	−1.27	1.37	−1.70	0.10	1.12	−0.40
	(−3.55)	(4.40)	(−4.27)	(0.26)	(3.56)	(−0.55)
RM VWAPs	−0.87	1.18	−1.10	−0.45	1.14	−0.10
	(−6.12)	(5.63)	(−2.70)	(−1.16)	(3.35)	(−0.13)

The piece's central argument is that the US dollar rises before foreign exchange fluctuations and falls subsequently. That holds true in various segments of the foreign exchange market and across different time periods and data sets. Not only does the discovery have enormous implications for academia, but there are many more. Asset managers, institutional investors, and businesses that receive US dollar cash flows and must convert them to their currency (or vice versa) should thoughtfully contemplate when to alter their portfolios due to the fact that the FX spot and derivatives markets are highly seasonal throughout the day. (Cerutti et al., 2024; Eichengreen et al., 2024)

IV. Reversal Returns

These terms characterise a trading strategy that capitalises on reversals by purchasing US dollars before to the Tokyo and ECB fixes and then selling them after the corresponding London and Tokyo fixes:

$$\Delta s_t^{\text{Tokyo}} = -\Delta s_t^{\text{pre-T}} + \Delta s_t^{\text{post-T}} \text{and} \Delta s_t^{\text{Europe}} = -\Delta s_t^{\text{pre-E}} + \Delta s_t^{\text{post-L}},$$

Where $\Delta s_t^{\text{pre-T}}$ and $\Delta s_t^{\text{post-T}}$ are the returns for the pre- and post-Tokyo periods, respectively. $s_t^{\text{pre-E}}$ and $\Delta s_t^{\text{post-L}}$and the London windows, which were in place before the ECB. Remember that you have the option to specify the reverse returns for the Tokyo window or the Europe window. Table I demonstrates that there was no apparent movement in either direction between the ECB benchmark and the London fix; therefore, the data eliminates that period.

A. Descriptive Statistics

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Table IV summarises the data for the DOL and the reversal returns of each currency. To start, we prove that daily reversal returns almost always have a positive bias. From 55% for the CHF to over 60% for the DOL, the discrepancies in reversals around the Tokyo benchmark are highly statistically significant for all currencies. In the European window, the Pacific and Asian currencies make up 51%, while the GBP, EUR, and USD each account for 55% or 56%. The discrepancies between the JPY and all other European currencies are quite substantial, with only the JPY having a p-value < 5%. In smooth and efficient markets, high-frequency returns are rare; thus, these reversal return figures are unexpected. Consequently, alterations significantly impact microstructure, as point-to-point consistently predicts reversals. This finding indicates a substantial impact on microstructure, as point-to-point consistently forecasts reversals. Among the currencies with daily close-to-close returns, only AUD has a positive return day more than 51% of the time.

Table IV. Statistical Properties of Return Reversals

In each panel, you can see the average, t-statistic, standard deviation, skewness, kurtosis, p-value, and the percentage of positive return data at the daily frequency. These are all calculated under the null hypothesis of a random walk. You may think of returns as BPS. Summaries of the Tokyo fix reversal and Europe fix reversal portfolios Panel A and Panel B display summaries of the Tokyo reversal portfolio and the Europe reversal portfolio, respectively. Panel A displays the Tokyo fix reversal portfolio, while Panel B displays the Europe fix reversal portfolio. Guidelines for doing a full U-turn are outlined in Appendix Note 2. The time span covered by the sample is from 1999 to 2019. Stats from Refinitiv's Tick History Database.

Panel A: Tokyo

Panel A: Tokyo	Frac >0	p-Val	Mean	t-Stat	Std. Dev.	Skew	Kurt
AUD	0.56	0.00	4.69	8.03	42.36	−0.33	11.00
CAD	0.57	0.00	3.02	10.68	20.52	0.77	17.50
CHF	0.55	0.00	2.49	7.19	25.10	0.80	35.93
EUR	0.57	0.00	4.14	12.60	23.85	0.25	9.10
GBP	0.56	0.00	3.15	9.87	23.17	−0.24	40.22
JPY	0.57	0.00	4.76	10.12	34.14	−0.05	16.12
NOK	0.58	0.00	4.89	11.93	29.75	0.47	18.59

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Panel A: Tokyo	Frac >0	p-Val	Mean	t-Stat	Std. Dev.	Skew	Kurt
NZD	0.56	0.00	5.68	9.29	44.37	0.02	7.77
SEK	0.56	0.00	3.96	9.56	30.05	2.30	58.38
DOL	0.59	0.00	4.09	14.46	20.50	0.02	8.88

Panel B: Europe

Panel B: Europe	Frac >0	p-Val	Mean	t-Stat	Std. Dev.	Skew	Kurt
AUD	0.51	0.04	2.34	3.20	53.11	0.14	10.93
CAD	0.52	0.00	2.31	4.01	41.76	0.36	7.76
CHF	0.54	0.00	4.79	6.81	51.09	−0.63	38.07
EUR	0.56	0.00	6.26	9.95	45.61	0.01	7.21
GBP	0.55	0.00	4.95	7.96	45.15	0.30	7.29
JPY	0.51	0.35	−0.12	−0.21	42.66	−0.18	7.13
NOK	0.52	0.00	3.54	4.51	56.96	0.03	6.31
NZD	0.51	0.04	2.16	2.70	57.95	0.08	8.64
SEK	0.55	0.00	5.68	7.25	56.81	−0.08	6.56
DOL	0.55	0.00	3.51	7.03	36.23	0.01	6.94

For context, consider that the S&P 500 index achieves positive daily returns 54% of the time. We find that returns around the Tokyo fix are positive in at least 70% of the sample months when we aggregate up the daily results to a monthly frequency. In the Europe window, positive reversal returns for European currencies occur at least two-thirds of the time. Nevertheless, this figure does not reflect the proportion of months that saw positive returns for the S&P 500 (for further details, see Section II of the Internet Appendix).

Although the cumulative effect of the continuous positive return bias is negligible prior to transaction costs, it produces substantial annualized returns thereafter. The annual return for the CHF and NZD in the Tokyo window might range from 6.2% to 14.2%. Table IV displays the daily results as a function of background points. The Europe window has very respectable yearly returns for the EUR (15.6%), GBP (12.4%), and CHF (12%). Annualized standard deviations of around 8%—roughly half of the standard deviation of the S&P 500 index throughout that time—occur whenever there is a turnaround, regardless of currency. Positive skewness and fat tails are common

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features of reversal return strategies. This phenomenon starkly contrasts with the negatively biased daily and monthly stock returns sampled during that time. The negatively biased daily and monthly stock returns sampled during that time starkly contrast with this phenomenon. during that time. Brunnermeier, Nagel, and Pedersen's carry portfolios and reverse portfolios are completely unique. Positive skewness and big tails characterize our reversal portfolios, which often provide positive returns. While carry trades have the potential to provide profits, they are very vulnerable to crash risk and characterized by large tails. It appears that interest rate disparities do not affect the skewness of returns in the same way as they do with close-to-close currency returns; in this case, low-interest-rate currencies are often associated with positive skewness. Finally, there are many positive aspects and excellent returns associated with reverse portfolios. The evolution of the reverse portfolios is the next topic we cover. (Gomez-Cram & Yaron, 2024)

B. Return Index Performance

Panels A and B of Figure 4 illustrate the daily reverse returns that we use to create return indices. When comparing the two time periods, we observe that reversal techniques are quite consistent. Any stock in the Europe window, with the exception of the JPY, will see its value rise over time due to the local fixings. There is a larger increase in value due to the local adjustments. A dollar invested in the Japanese yen would have been worth almost $12.20 by the end of 2019 if traders had adhered to the Tokyo fix. The ultimate portfolio values for the EUR, GBP, and DOL are 8.9, 5.3, and 8.6 US dollars, respectively, when the same procedure is used. However, the value of the JPY portfolio drops by almost 6%. In the Europe window, the prices for the EUR, GBP, and DOL portfolios are 26.9, 13.5, and 6.4 US dollars, respectively.

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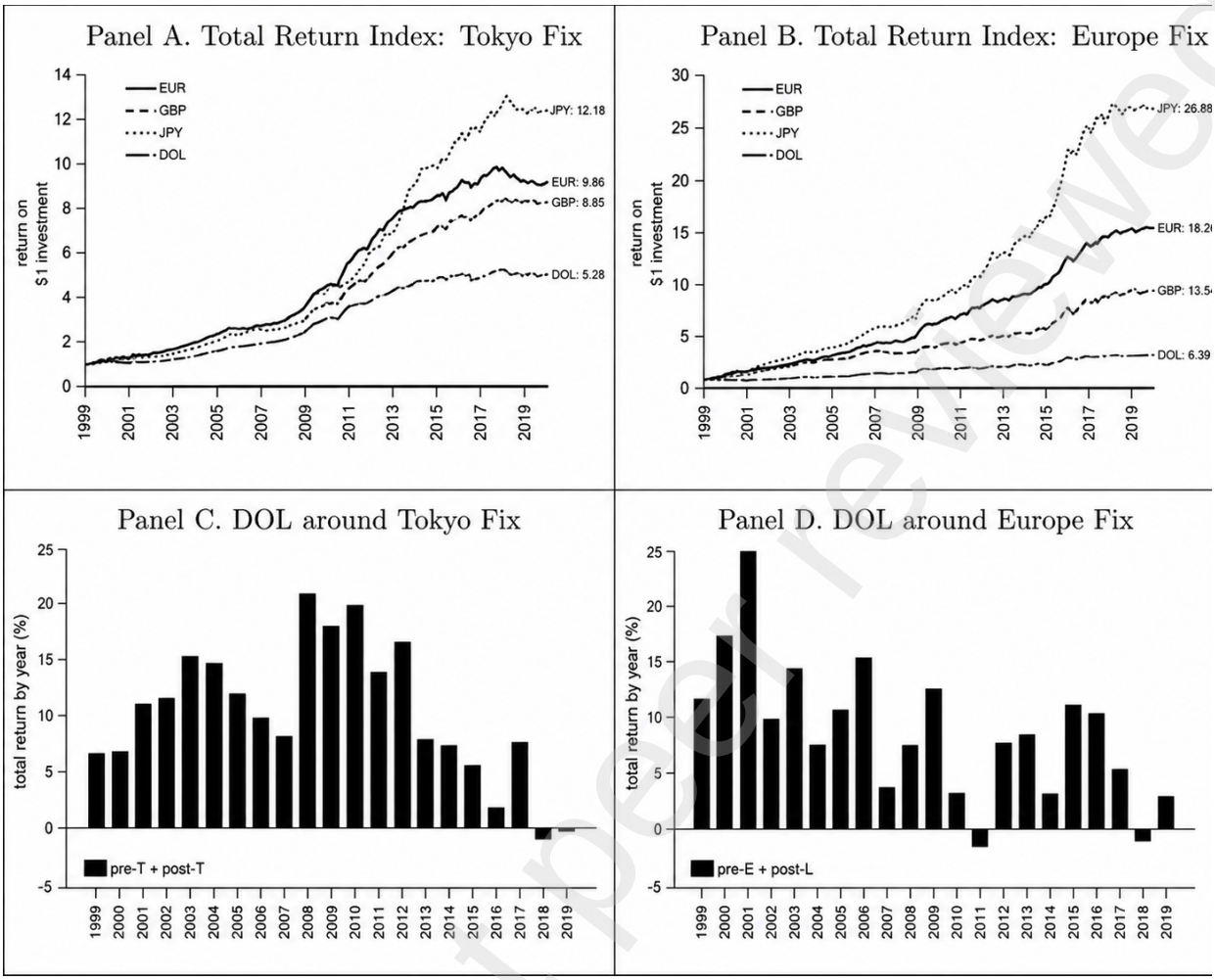


Figure 4: Success in trading from year to year and total return measurements vs. the results of a trading plan that took into account the shifts in Tokyo and Europe are shown in the panels. An investor employs this tactic while purchasing US dollars at a prefix and selling them at a postfix. With a one-dollar initial investment, the following are the total return indices for the dollar portfolio (DOL), together with the three major currencies (EUR, GBP, and JPY). The annual profit of the dollar portfolio is displayed in the lower panels. The lower panels display the annual profit of the dollar portfolio. in the lower panels. The lower panels display the annual profit of the dollar portfolio. The lower panels display the annual profit of the dollar portfolio. The lower panels display the annual profit of the dollar portfolio. See Appendix Note 2 for information on reversal tactics. The research period begins in January 1999 and ends in December 2019. The data originates from Refinitiv Tick Analysis. There is a colored figurine available at

342

343 Figure 4's panels C and D show the DOL's reversal returns year-over-year. During the time of the
344 economic crisis and the dot-com bubble bust in 2001, the returns for both approaches were over
345 20% annually. Returns were quite low during the 2007 European and Tokyo windows.

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Typically, reversal returns are positive, but there is a downward trend at the end of the data period, and the cause remains uncertain. It's anyone's guess. First, there is anecdotal evidence that the predictability of intraday returns is decreasing, and arbitrage capital has been capitalizing on these reversal patterns since the later part of the sample period.

Second, after the 2013 WM/Reuters rigging incident, several regulatory adjustments were put into place. The lower reversal returns and lower average trading volumes around the fixes in the most recent data period could be causing these disparities. Nevertheless, there is little evidence to implicate the rigging incident or the subsequent changes in the variations in the severity of reversal returns. Actually, average reversal returns in Europe were among the greatest in 2015 and 2016. Overall, the data for all nine G9 currencies show reversals around the fixes during the long study period. (Krohn et al., 2024; Chortareas & Noikokyris, 2024)

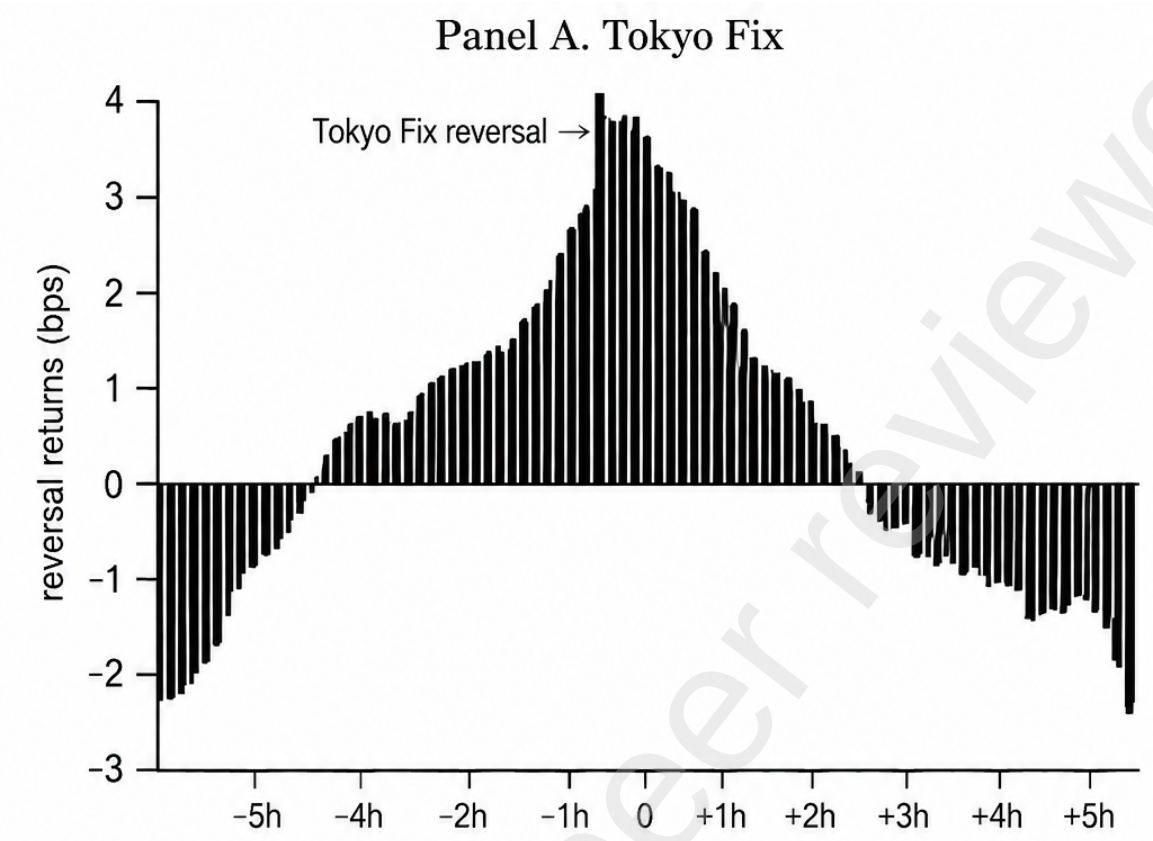
C. Alternative Reversal Timing

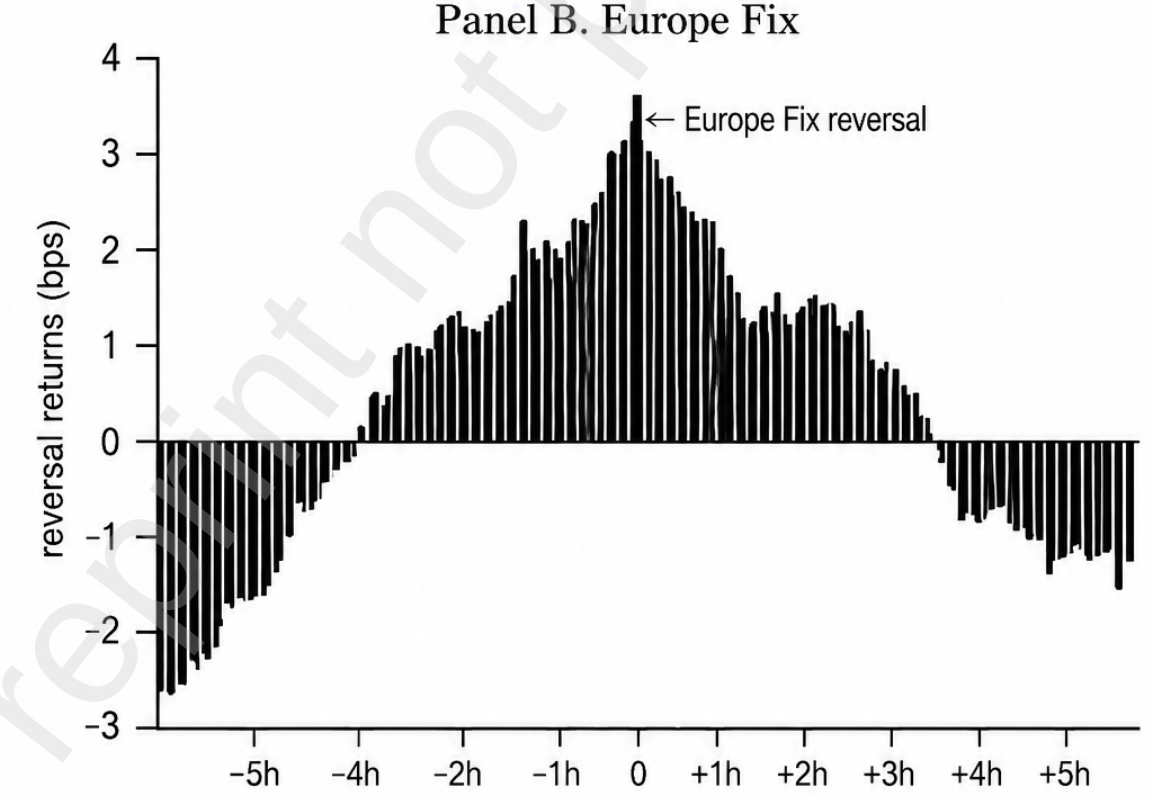
The next step is to determine if the fixes indicate the major turning moments of the day or whether the reversals seen are a more common occurrence in the data, resulting from return autocorrelation rather than benchmarks.

The reversal points are shifted away from their respective fixes, and separate reversal returns for the European and Tokyo windows are calculated for this purpose. You may see the results of shifting the reversing points five minutes either way in Figure 5. Our normal reverse return is the black bar in each section of the two windows. The other bars show windows of the same length that have been shifted up or down the timeline as they play again.

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Figure 5: There are instances when reversal occurs. The graphic shows the reversal outcomes for various reversal periods in Tokyo (Panel A) and Europe (Panel B) during the major trading hours. Basis points (bps) represent returns on the y-axis. The time of turnaround relative to Tokyo and European fixes is shown by the black bar and the number "0" on the x-axis. The black bars represent the repair, and the values to the left and right of them indicate when the fix is reversed. Note 2 in the appendix also discusses reversal approaches. This cohort begins in January 1999 and continues through December 2019. Results from the Refinitiv Tick historical database.

When the reversal points coincide with the fixes for the Tokyo and Europe windows, we see that the reverse return is optimal. The average returns are reduced when the reversal time is subtracted from the adjustment, and they continue to go down as the adjustment moves farther away. Daily return reversals are therefore not only a common occurrence in the data, but they also occur at the same time as the announcement of FX standard rates. The next section provides more evidence in support of this viewpoint.

V. Natural Experiments

In Section 3, we see how the fixes are crucial for finding intraday reversals in three quasi-natural experiments that take advantage of (i) Japan's failure to follow DST, (ii) the fact that Japanese and American holidays are not synchronous, and (iii) data taken before and after the 1994 and 1999 ECB and London fixes, respectively.

A. Daylight Saving Time (DST) Analysis

The time difference between Tokyo and New York City fluctuates throughout the year since Japan does not observe Daylight Saving Time. Over the winter, it drops by 14 hours, and between March and November, it drops by only one. The movement of the EUR, GBP, and JPY during the Japanese trading hours, which are connected to ET (New York time), is seen in Figure 6. The constant line represents the return trends throughout winter. However, during US Daylight Savings Time, the dotted line displays the return patterns. For about eight months of the year, the data clearly show that the appreciation of the US dollar at the Tokyo fix does not occur simultaneously. The lack of a pattern associated with the beginning or ending of the workday is, nonetheless, obvious. While this lack of a pattern does not affect the movement associated with the opening

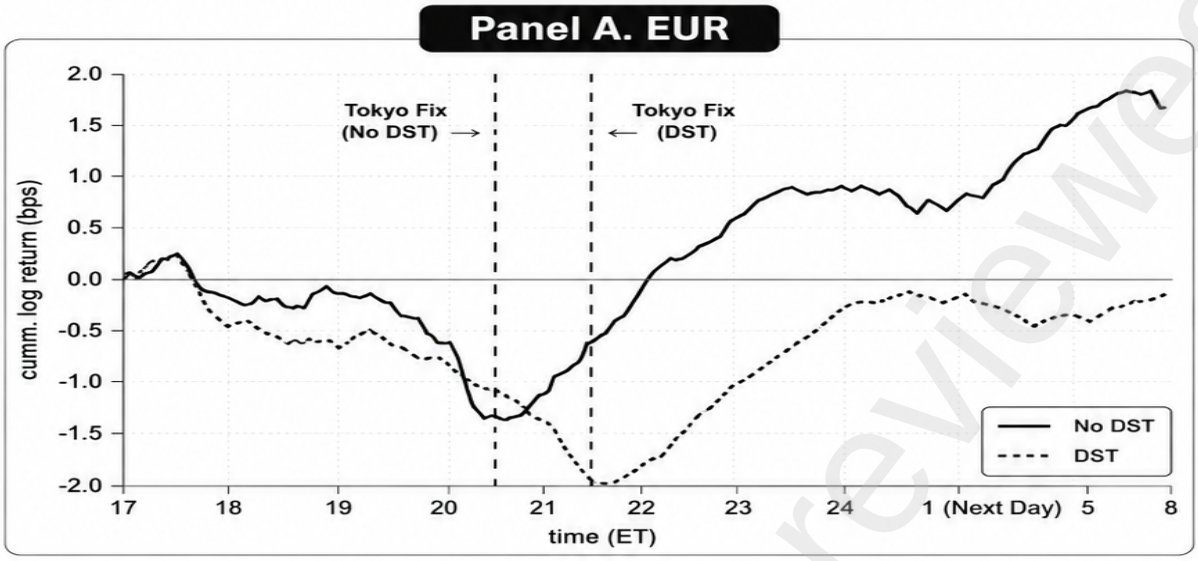
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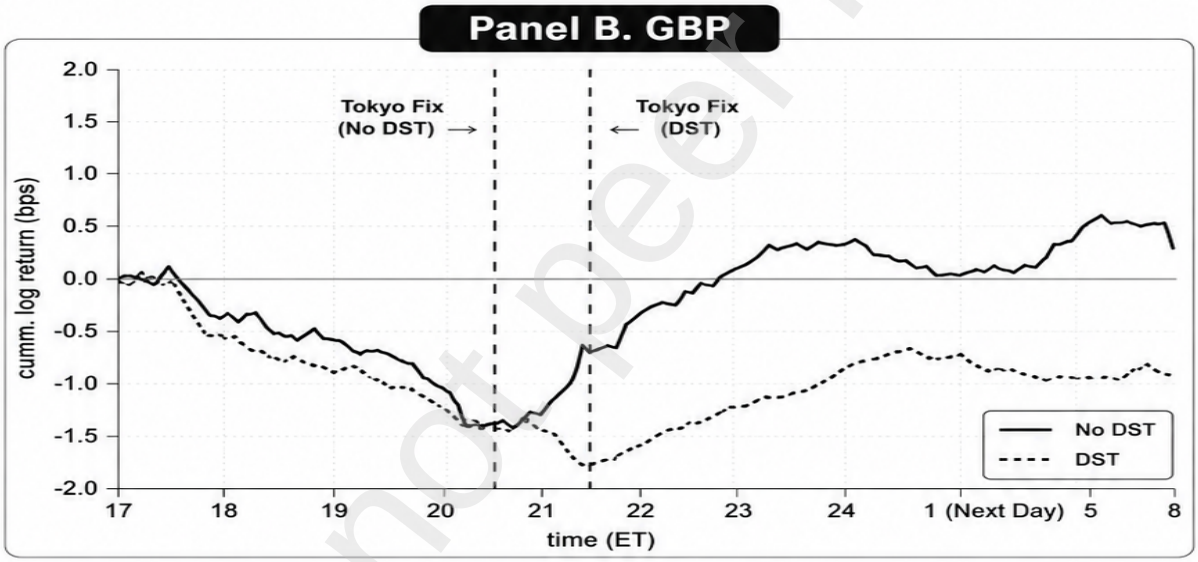
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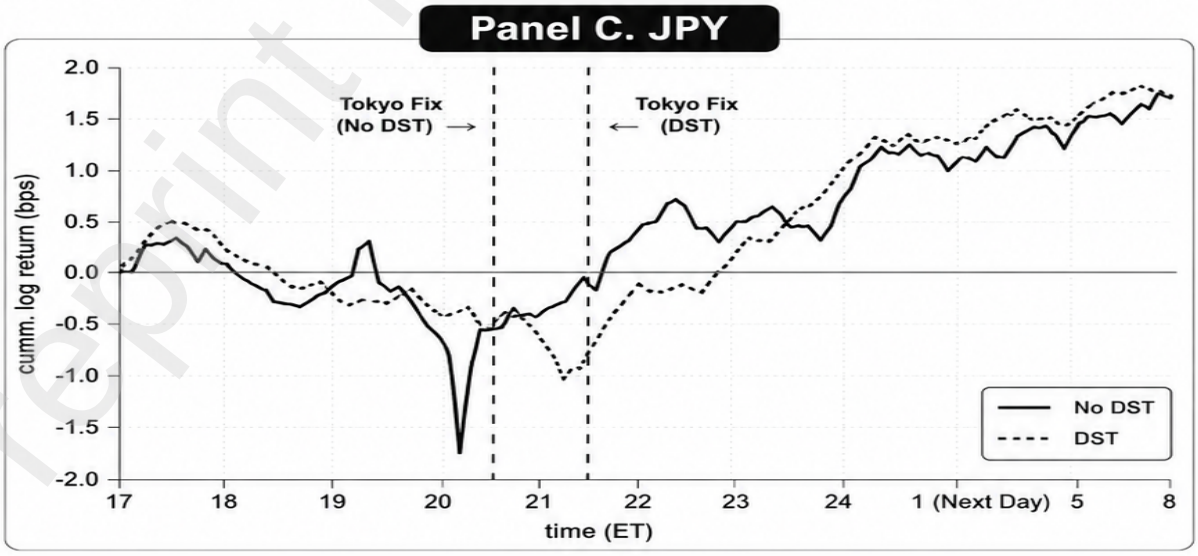
and closing times of the Asian market in Eastern Time, it does indicate a significant shift in the spike of the price pattern associated with the Tokyo fix. (Curcuru et al., 2024)

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Figure 6: Intraday results are affected by daylight savings time changes. This chart shows the cumulative average 5-minute return. Δs Panel A shows the EUR/USD price, Panel B shows the GBP price, and Panel C shows the JPY price during the major trading hours in Japan. The units used are basis points (bps). On days when New York's time zone is in Eastern Standard Time (dotted line), prices Prices are displayed on days when New York's time zone is in Eastern Standard Time (dotted line).. On days when it is in Eastern Daylight Saving Time (solid line), prices are given for days without DST. The value of the US dollar declines when other currencies appreciate. At 19:55 and 20:55, the two dotted lines represent the Tokyo fixed time in all time zones. This cohort begins in January 1999 and continues through December 2019. Final tick records are indicated by the color chart.

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391 **B. Asynchronous Holiday Effects on Financial Markets**

392 Japan offers a second, nearly natural experiment by taking advantage of the fact that international
393 holidays do not always coincide. Our sample excludes 304 Japanese holidays that are not
394 recognised in the United States. To test if the adjustments really do produce daily reversals, we
395 may observe whether the reversals discussed in Section IV occur on holidays because Tokyo fixed
396 reference rates are not issued at those times.

397 Table V displays the average daily returns in bps for all 4,958 days of our data collection, including
398 the 304 Japanese holidays. Table V presents the average daily returns in basis points (bps) for all
399 4,958 days of our data collection, including the 304 Japanese holidays. Transaction fees are not
400 included in these returns. All of the reversal returns on regular business days are substantial and
401 very statistically significant, as seen in Table IV. The findings disappear during Japanese holidays
402 because the average returns are much lower and statistically insignificant. Several currencies see
403 negative average returns, with the JPY and DOL seeing drops of almost 75% and 80%,
404 respectively. Despite far fewer data, we observe that the magnitudes are significantly greater on
405 Japanese holidays and statistically extremely significant for all European currencies and the DOL
406 when we apply the same split for the reversals around the European fixes. In the third and sixth
407 columns of Table V, we provide the p-values from a one-sided test that compares Japanese holiday
408 reversal returns to those on business days, with the alternative that they are lower. We reject the

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null for almost all coins in the Tokyo session at the 5% threshold. Unfortunately, the null for the Europe window cannot be turned down.

Table V. Reversal Returns: Japanese Holidays and Business Days

For Japanese holidays ("JP H-Days", 304 days) and business days ("JP B-Days", 4,958 days) when shorting foreign currencies during the prefix period and desiring them during the postfix period, the average reversal returns in basis points are shown in this table. The t-statistics are also shown. A one-sided t-test disproves the null hypothesis that JP H-Day returns are lower than JP B-Day returns. H_A:JP H - Days < JP B - Days related to the Tokyo-Europe repair. The time frame of the sample is from January 1999 to December 2019. Stats from Refinitiv's Tick History Database.

Variable	Tokyo JP H-Days	JP B-Days	p-Val	Europe JP H-Days	JP B-Days	p-Val
AUD	-1.10	4.92	0.01	5.12	2.28	0.80
CAD	1.74	3.24	0.12	3.86	2.30	0.76
CHF	0.18	2.74	0.07	5.75	4.56	0.64
EUR	2.38	4.43	0.07	8.61	5.88	0.81
GBP	1.53	3.39	0.11	8.64	4.93	0.91
JPY	1.38	4.81	0.02	1.48	-0.20	0.75
NOK	1.65	5.22	0.03	4.06	3.37	0.56
NZD	0.48	4.86	0.03	5.72	2.01	0.87
SEK	0.07	4.05	0.02	7.94	5.31	0.77
DOL	0.93	4.12	0.01	5.60	3.23	0.85

We discussed the significance of time for the reversals around the Tokyo benchmark in the previous section. But this section's data shows that reversals are rare on days without a benchmark rate. Reversal returns during the Europe window are statistically similar to each other since both the US and Europe trade on the same days. Finally, it's obvious that return reversals on the Tokyo trading day were significantly impacted by the Tokyo benchmark. (Degiannakis et al., 2024; Fasanya et al., 2024)

C. Return Dynamics Following the Introduction of Currency Benchmarks

The ECB adjustment will have a different release window in 2016. Our primary data set includes all three of the aforementioned FX metrics from the inception of the Euro in 1999 up to the present day. As a final, seemingly logical experiment, we may compare the pre- and post-change patterns

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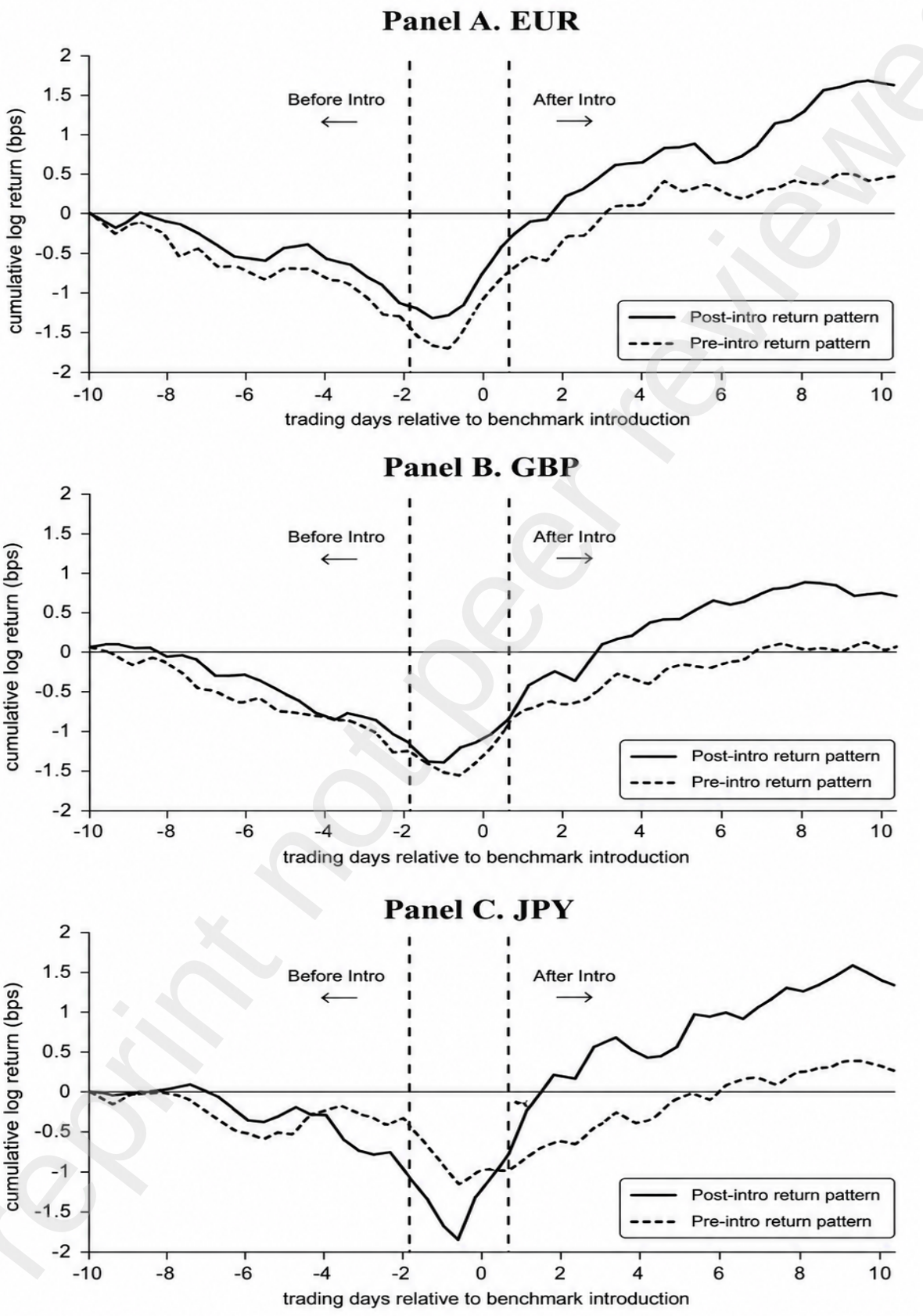
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of intraday currency returns. We can do so by extending our sample period all the way to the mid-1980s.

In 1994, the London benchmark was first proposed (Melvin and Prins, 2015). Figure 7 shows the daily return trend from February 1986 to December 1993 (the dotted line) and from January 1999 to December 2019 (our reference period, the solid line). According to the data, the first hour of trading, which begins at around 8:00 ET and concludes with the application of the London fix, is when the value of foreign currencies rises. Afterwards, the German mark continues to rise until the conclusion of the German business day. But during the whole of the New York trading session, the pound rises. The yen's value rises first before falling.

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Figure 7: Intraday returns dynamics: Introducing the London fix. This chart shows cumulative average 5 minutes returns Δs Panel A shows the EUR, Panel B shows the GBP, and Panel C shows the JPY. These prices were measured in basis points (bps) during the busiest trade hours in London and New York. Before the London fix was put in place, from February 1986 to December 1993, the dotted red line shows the time. Between January 1999 and December 2019, the solid blue line shows the time. The two faint lines across the graph show the ECB fix (8:15 ET) and the London fix (11:00 ET). They got the data from Olsen Financial Technologies for the years before 1999;

437

438 We find it intriguing that the patterns for our benchmark sample period change drastically after
439 the London adjustment. The patterns for our benchmark sample period change drastically after
440 applying the London adjustment, which we find intriguing. All three currencies: This morning in
441 New York, all three currencies continue to rise, with the ECB adjustment at 8:15 ET marking the
442 tipping point. In New York early this morning, the ECB fix at 8:15 ET was seen as the tipping
443 point. All currencies see a little decline prior to the London fix. Following the correction, the Euro
444 and the British Pound reverse direction and continue to rise until the trading day comes to a close.
445 The ECB fix, published in 1999, will be discussed next. In July 2016, the European Central Bank
446 began to postpone the revelation of the reference rates until 16:00 CET to discourage clients from
447 using the ECB benchmark in their trading. For the years before the introduction of the EUR (Panel
448 A); from January 2000 to June 2016 (Panel B); and after the change in release time (Panel C),
449 Table VI illustrates 15-minute periods around the ECB fix. A shift in currency patterns The first
450 portion of the data prior to the ECB fix reveals a shift in currency patterns. in the first portion of
451 the data prior to the ECB fix. About 7.15 and 8.15 were when the change occurred. Before 7:15
452 a.m., the three currency pairings often have negative returns, but these are usually not statistically
453 significant. At 8:15, they have turned positive, which is a favorable indication. It is common for
454 foreign currencies to have significant gains during the first hours of trading in New York, as
455 previously observed. The precise moment of transition, however, is difficult to ascertain. Since the
456 ECB intervention was implemented, that has changed. The data in Panel B demonstrates that the
457 15-minute intervals beginning at 7:45 and ending at 8:00 often have negative returns that are
458 substantial. Afterwards, it is evident that the situation reversed. Between 8:15 and 8:30, there were

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significant and strong advances for the EUR and the GBP. Most striking are the findings shown in Panel C. The revised WM/Reuters 14:00 CET benchmark, which accounts for the shift in when the ECB announces its reference rate and shifts the turning point to 8:00 ET, is the source of these figures. The 7:45–8:00 time slot has experienced negative returns since July 2016. On the other hand, from 8:00 to 8:15 at the new postfix time, all three currencies show positive and statistically significant results. (Filippou et al., 2025)

Table VI. Introduction and Changes to ECB Fix

The tables below show the 15-minute returns in basis points before and after the ECB adjustment at 8:15 ET. Panel A shows the years before the ECB fix, which run from February 1986 to December 1998. Panel B shows the years from January 2000 to June 2016, when the ECB fixed reference rates were released at 8:15 ET. Panel C displays the years from July 2016 to December 2019, when the London fix rates were released later. The t-statistics in the boxes come from Olsen Financial Technologies and Refinitiv Tick History.

Panel A:

Variable	07:45–08:00	08:00–08:15	08:15–08:30
EUR	−0.06	0.53	0.23
	(−0.50)	*(4.02)*	*(1.54)*
GBP	0.13	0.22	0.06
	(1.18)	*(1.81)*	*(0.36)*
JPY	0.26	0.44	0.35
	(2.02)	*(2.58)*	*(2.35)*

Panel B:

Variable	07:45–08:00	08:00–08:15	08:15–08:30
EUR	−0.58	−0.93	1.01
	(−4.59)	*(−7.58)*	*(7.76)*
GBP	−0.40	−0.20	0.52
	(−3.78)	*(−1.83)*	*(4.36)*
JPY	−0.15	−0.34	0.21
	(−1.45)	*(−2.97)*	*(1.60)*

Panel C:

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Variable	07:45–08:00	08:00–08:15	08:15–08:30
EUR	−1.28	1.09	−0.08
	(−5.89)	*(6.18)*	*(−0.46)*
GBP	−0.35	1.02	−0.52
	(−1.56)	*(3.84)*	*(−2.33)*
JPY	−0.38	0.54	−0.40
	(−2.45)	*(3.41)*	*(−1.98)*

475

476 **D. Evolution of Trading Patterns Around the London Fix**

477 The high-frequency flow data does not account for the period prior to the modifications. Instead,
478 we display the volume of trade based on the number of changes in CME FX futures quotations,
479 which we receive every five minutes. every five minutes. From 8:35 ET to 14:00 ET (the time of
480 the London fix), Figure 2 illustrates the average number of updates in the JPY and GBP quotations
481 relative to the total number of changes.

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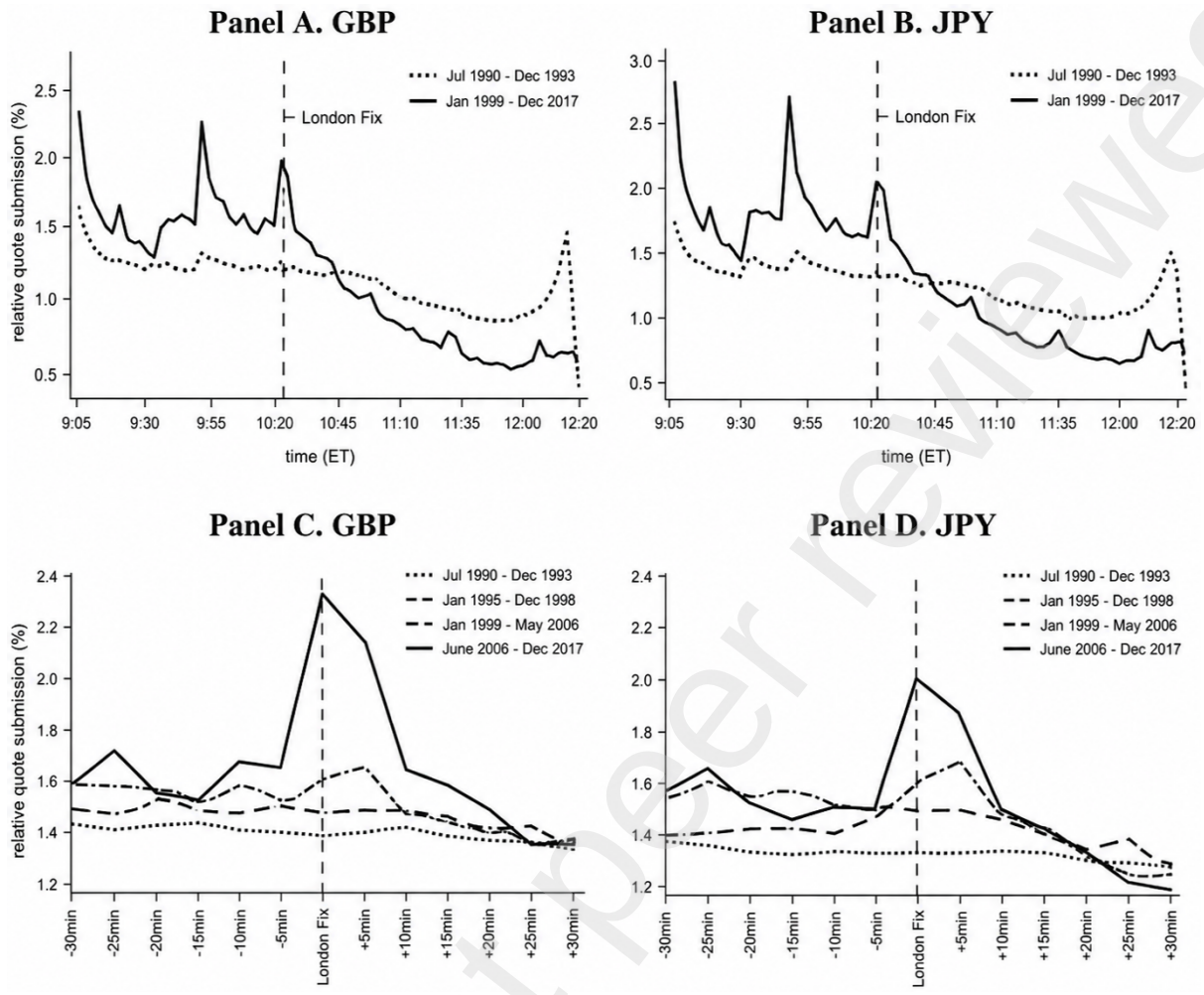


Figure 8: Variations in the volume of trades occurring around the London fix. Compare the number of quotes delivered daily between 8:30 AM and 2:30 PM ET for GBP and JPY to the number of changes to quotations in CME-traded currency futures that occur every five minutes. The graph demonstrates the difference. Figures A and B, located in the top row, display the changes in the futures market from 1980 to 1993 (dotted line) and from 1999 to 2017 (solid line). The following lines illustrate the activity of quotations twenty minutes before and after the London fix: (dotted line) July 1980–December 1993 (dash line) January 1995–December 1998 (dash-dotted line) January 1999–May 2006, and (solid line) June 2006–December 2017. This originates with CME. There is a colored figurine available at

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483 The dotted line in Panels A and B represents the busiest periods in New York prior to the
484 implementation of the patch. The trade hours in the morning and afternoon were as follows. Aside

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from that, the line remains rather level, with a slight increase around 10:00 ET, the coin options' expiration time. From January 1999 to October 2021, the pattern is different when we examine our baseline group (the continuous line in Panels A and B). At the London benchmark, two currencies both move up. Above everything else, keep this information in mind. Around 10:00 ET, there is a larger increase at the same time. There is still a lot more activity even when currency options finish early in the morning. On the other hand, since the London benchmark was implemented, there has been far less activity in New York during the late trading hours.

We use the one-hour window in Panels C and D at around 11:00 ET on various occasions. There was obviously no haste in the dealings that occurred approximately 11:00 ET before the distribution of the update (dotted line). Furthermore, the increase becomes much more apparent in the years following 2006. For the two times after the repair, you may see the gradual increase in quote activity beginning at around 11:00 ET.

When comparing trader behavior before and after the London patch, a noticeable change occurs. When comparing trader behavior before and after the complete implementation of the London patch, there is a noticeable change. While the findings in this section explore the study's central argument and touch on the concept of "liquidity breeds liquidity," they do not directly support it. By facilitating the discovery of dealers and clients and the execution of mutually beneficial trades, benchmarks (or fixes) enhance social surplus, as shown by Duffie, Dworczak, and Zhu. (Karnaukh et al., 2025)

VI. Order Imbalances and Price Reversals

This section argues that the literature on the microstructure of the foreign exchange market can provide insight into the return patterns seen from an economic perspective. Standard wisdom is that (i) varying degrees of market knowledge and (ii) supply and demand management by market participants are the two primary drivers of sudden price shifts. The second school of thought is that arbitreurs, who act as financial intermediaries, should get a premium for taking on inventory risk in exchange for liquidity. Liquidity providers need the bid-ask spread and the estimated return on hazardous assets in order to get further payment. A temporary shift in supply and demand dictates a new path for pricing in this scenario. We show that our primary empirical result is compatible with an explanation predicated on an inherent need for instantaneous U.S. currency at

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the fixes. Numerous other plausible explanations, such as recent developments in monetary policy and the economy, are investigated and rejected. This indicates that (unequal) information is insufficient to explain the observed daily return patterns. Volatility, price surges, or shifts in market liquidity during the day cannot account for the patterns. (Hancu & Ranaldo, 2025; Pasquariello, 2024)

A. Inventory Risk Management

Sellers often make available liquidity to traders in a timely manner using conventional inventory management techniques. To provide space for new merchants who arrive later in the day, the dealers sell what they have. Prices in these models shift directions during "liquidity events," which middlemen trigger by handling order imbalances. When order mismatches occur or liquidity events occur throughout certain periods of the day, we say that these are the major fixes. We examine the volume of D2D transactions on RM, the primary interdealer platform for Commonwealth currencies, to provide evidence in support of this claim. Figure 9 displays the daily trading volume for the G9 currencies. Over the course of a day, we calculate the average volume transferred each hour. During the two hours around the London fix, the volume of currency trades peaks (17.5% to 27%). The hours leading up to the ECB fix are the second busiest. European and US trade hours see the bulk of RM trading activity. On the other hand, the Tokyo fix for the NZD, AUD, and JPY displays greater movement on the heatmap.

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Time (ET)	USD	EUR	JPY	GBP	AUD	NZD	CAD	CHF	SEK	NOK
17:00 - 18:00	1.03	2.00	0.62	0.36	0.64	0.46	0.81	1.36	2.51	1.51
18:00 - 19:00	1.61	2.96	0.30	0.17	0.29	0.29	1.20	2.07	2.21	1.85
19:00 - 20:00	2.39	3.86	0.38	0.21	0.36	0.28	2.13	2.63	3.28	2.38
20:00 - 21:00	4.71	5.91	1.00	0.57	1.02	1.04	3.60	4.02	4.94	3.48
21:00 - 22:00	5.15	6.09	1.20	0.82	1.11	1.01	3.96	5.62	4.51	3.54
22:00 - 23:00	3.71	3.18	1.28	0.64	0.95	1.05	2.81	3.50	3.86	2.52
23:00 - 00:00	3.20	2.71	1.08	0.47	0.77	0.87	2.81	2.47	2.36	2.18
00:00 - 01:00	2.76	2.67	1.09	0.51	0.84	0.98	2.60	2.86	2.80	2.40
01:00 - 02:00	3.44	3.21	1.38	0.66	1.11	1.38	3.60	3.77	3.88	3.63
02:00 - 03:00	5.68	5.66	2.18	1.19	1.98	1.92	4.12	4.66	4.26	4.26
03:00 - 04:00	6.29	5.36	2.57	1.30	2.30	2.17	4.75	5.40	5.32	5.19
04:00 - 05:00	6.73	5.62	2.62	1.28	2.43	2.58	5.02	6.03	5.22	5.43
05:00 - 06:00	4.33	3.81	2.00	1.10	2.02	2.04	3.87	4.61	3.72	3.86
06:00 - 07:00	3.17	2.66	1.62	1.01	1.74	1.66	3.47	3.51	3.31	3.09
07:00 - 08:00	7.34	9.35	6.17	4.37	6.63	6.11	6.21	6.81	6.18	6.10
08:00 - 09:00	11.06	13.85	9.92	7.29	8.18	8.13	9.12	12.07	7.51	10.19
09:00 - 10:00	9.43	11.61	11.29	11.08	13.11	12.39	10.57	14.64	10.11	17.60
10:00 - 11:00	6.10	6.56	7.04	7.54	7.97	7.73	6.72	7.88	5.95	7.63
11:00 - 12:00	3.23	3.98	3.14	2.30	2.54	2.62	4.02	4.43	3.38	4.04
12:00 - 13:00	2.57	3.64	1.94	1.17	1.63	1.69	2.93	3.18	2.81	3.15
13:00 - 14:00	2.82	3.14	2.16	1.36	1.83	1.87	2.87	2.66	2.66	2.87
14:00 - 15:00	2.98	2.86	2.28	1.86	2.03	2.09	2.91	2.56	2.36	2.45
15:00 - 16:00	1.90	2.08	1.62	1.10	1.43	1.36	1.86	1.82	1.68	1.86
16:00 - 17:00	1.24	2.77	1.67	0.89	1.17	1.08	1.78	1.09	1.34	1.16
	NY Close	London Close	Tokyo Fix		Tokyo Fix	Tokyo Fix	London Fix	ECB Fix	NY Open	NY Open

Figure 9: One day's trading volume compared to the average hourly volume exchanged during the trading day, the graph displays the percentage of the day's total volume that was traded. Your total is 100 because of all the boxes. Data collection began in June 2006 and will conclude in December 2019. Refinitiv FX Matching is the source of the data. There is a colour figurine available at

532

533 To compensate traders for the liquidity they provide, inventory risk models offer them money in
534 the form of bid-ask spreads and positive predicted returns on hazardous products. They achieve
535 this by accumulating US dollars prior to the repair and reselling them to customers at a premium
536 during the fix. This strategy works so long as the return patterns demonstrate a structural need for
537 US dollars during the repair. When dealing with currency exchange, the strategy is known as
538 "prefix hedging." Actually, banks that are cognizant of order imbalances express a clear desire to
539 safeguard their positions, which may involve delaying the fulfillment rates of client orders.
540 In the following step, we will use RM's volume and order flow data to verify some of the most
541 fundamental claims made by inventory management models. When compared to previous

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research, our focus is on the British pound and the Australian dollar in relation to the European euro. This discrepancy is due to two primary causes. First, RM only applies to Commonwealth currencies, so EUR and JPY volumes on RM aren't always accurate. Second, during US and European business hours, there is a significant increase in RM trading. We will discuss findings that are comparable to those shown in the Internet Appendix for the Tokyo benchmark. (Evans & Rime, 2024)

B. Order Flow Dynamics and Market Imbalance

The relationship between unconditional reversal patterns and unconditional order imbalances is examined first. According to Grossman and Miller, if order flows are normal, reverse returns should be zero. Returns on reversals are non-zero because market makers have a limited risk tolerance and these flows occur at different times. During fixing periods, the US dollar supposedly reaches a local high. This indicates that at every adjustment there is either a net supply of foreign currency or an unconditional net demand for US dollars.

Table VII, Panel A summary of the RM order flow before and after the change. The order flow is the sum of all deals initiated by buyers and sellers minus one another. That is why the presence of negative order flow indicates a desire to purchase US dollars. We provide both the mean and median, but we will focus on the median due to the skewed distribution caused by large outliers. Both currencies' medians and means are positive after the London fix. They were pessimistic prior to the repair. Equally crucial from a statistical perspective are the prefix and postfix windows' medians. Each currency's prefix window has an order imbalance of about $20 million. As a percentage of total trading activity at the London fix, the order imbalance accounts for around 6.5 percent for GBP and 7.5 percent for AUD. It further reveals that both currencies have a lot more days with negative (positive) order flow before (after) the London benchmark than with positive (negative) order flow. Looking at the ECB benchmark, which occurs around three hours before the London fix, yields almost identical prefix outcomes. Our research shows that trades centered on the London benchmark are not the only source of the unconditional orders discrepancy.

Table VII. Summary Statistics: Order Flow around European Fixes

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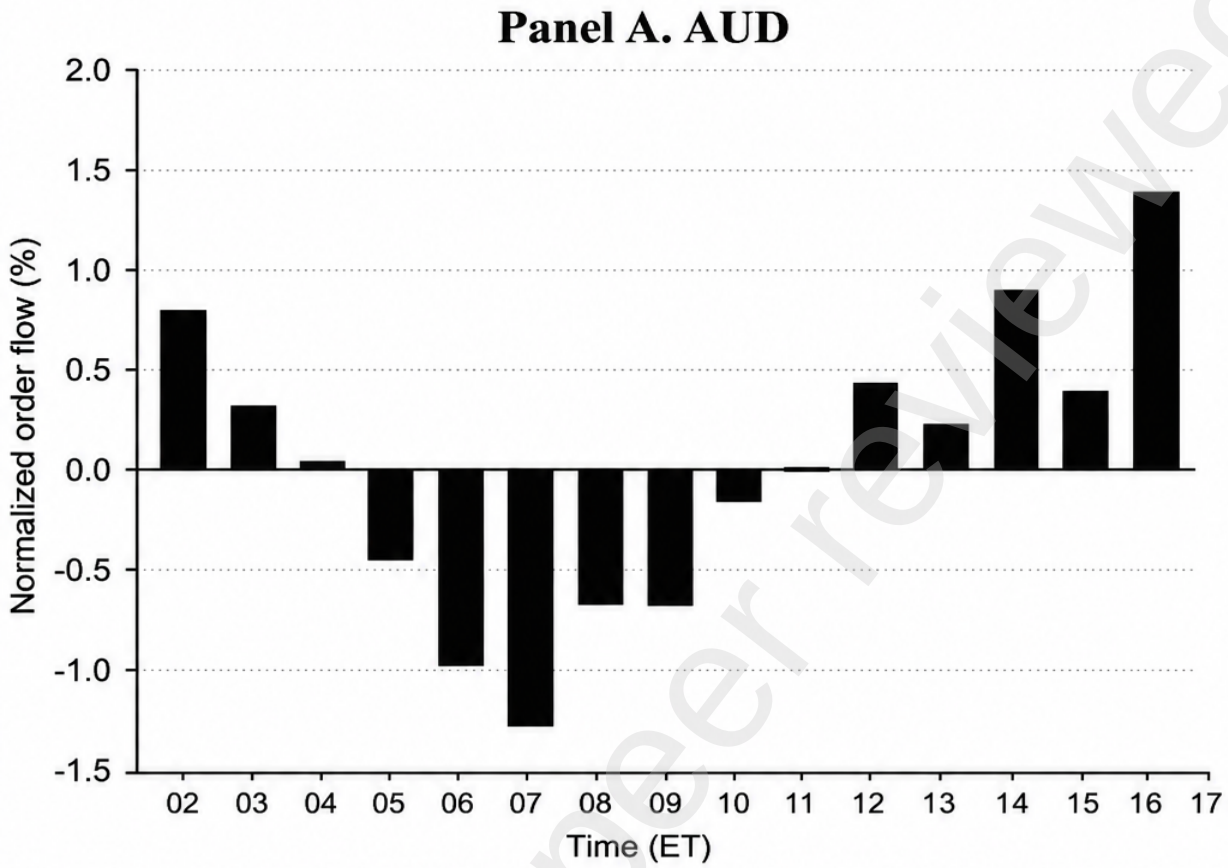
Each day's mean, median, z-statistic, standard deviation, skewness, and kurtosis, together with the proportion of observations demonstrating positive order flow, are shown in Table 1. Under the null hypothesis of a random walk, the p-value is derived from a two-sided test of detecting one-way order flow. In statistics, the z-statistic is synonymous with testing the hypothesis that the median does not equal zero. The order flow, expressed in millions of dollars, is the volume differential between the volume started by buyers and sellers. Refer to Note 1 for more details about the significance of the intraday timings. June 2006 to December 2019 is the time span of the sample. You may locate it at Refinitiv FX Matching.

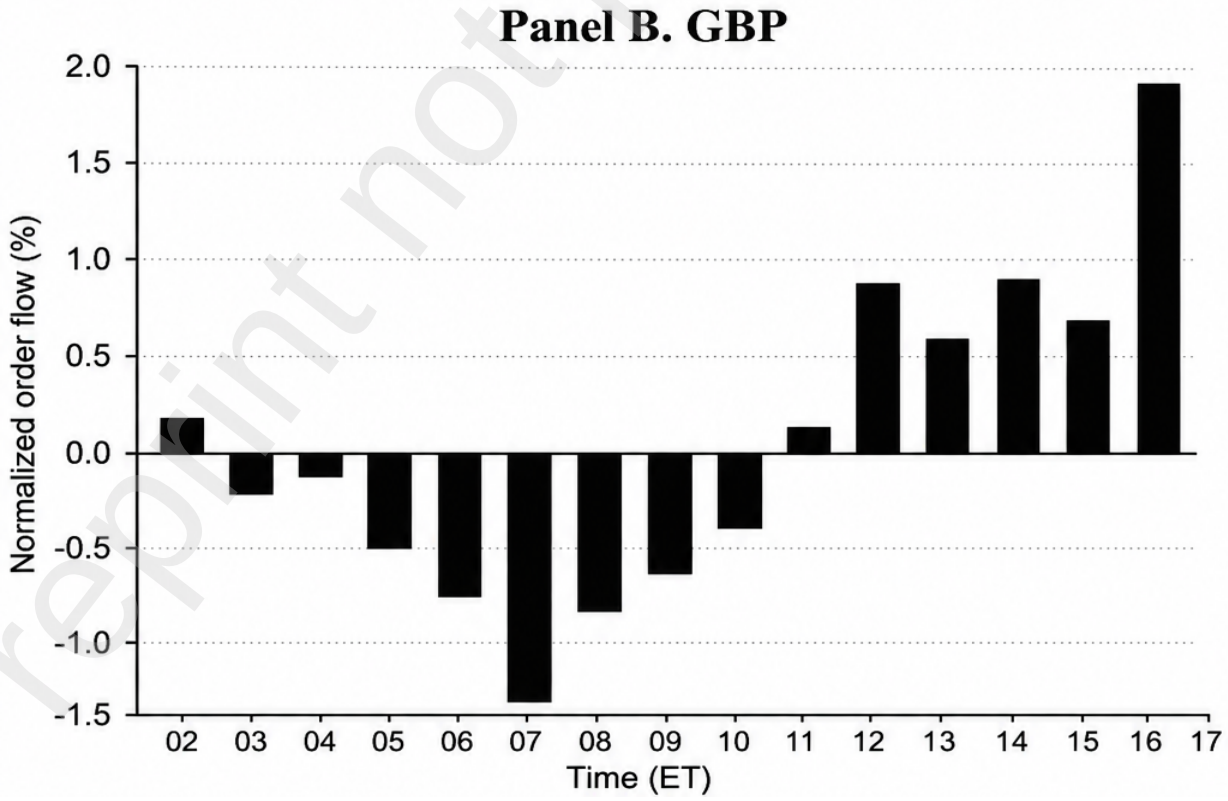
	Pre-E	**Pre-L**	**Post-L**	**Pre-E**	**Pre-L**	**Post-L**
Fraction Positive	0.47	0.46	0.53	0.47	0.47	0.55
p-Value	0.02	0.01	0.02	0.03	0.03	0.01
Mean	−8.21	−25.32	2.64	2.46	−3.56	21.68
t-Statistic	(−1.94)	(−4.11)	(0.74)	(0.32)	(−0.35)	(4.91)
Median	−10.58	−20.11	6.54	−13.48	−21.20	17.92
z-Statistic	(−2.85)	(−3.56)	(2.63)	(−2.37)	(−2.14)	(4.67)
Std. Dev.	241.84	355.28	214.17	438.96	604.83	268.44
Skewness	−0.22	−0.24	−0.44	0.59	0.35	0.19
Kurtosis	6.18	5.48	9.13	10.19	7.32	7.84

You can observe the average normalised order flow hourly throughout the day in Figure 10. People are purchasing with US dollars when the bars are negative (grey) and selling when they are positive (black). The graphic depicts the transformation of historical pricing patterns into prospective quantity patterns in the interdealer market. It reveals that demand for dollars is high before the London benchmark but that demand decreases thereafter.

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Figure 10: What we call "normalised order flow" is really just the sum of all transactions minus the sum of all deals begun by sellers divided by the total volume of trades. June 2006 to December 2019 is the time span of the sample. You may find it at Refinitiv FX Matching.

Here we lay out the rationale for the D2C company's requirement for US cash. However, our order flow data is exclusive to D2D systems. After we determine the underlying demand for US dollars from our data, we believe that D2D order flow will be a suitable representation of D2C order flow. According to a large body of literature on asymmetric information and inventory models, D2D order flow ought to account for both pre- and post-event pricing variations. The fact that there are two tiers to the foreign exchange market implies that a few liquidity providers process the vast majority of deals. In simple terms, a large number of consumers are served by a small number of vendors. To mitigate risk, these vendors deal directly with buyers and aim to maintain a low dollar supply or demand. If leftover imbalances can't be hedged where they occur, they go where they occur; they are forwarded to interdealer platforms, where other liquidity traders buy and sell them. Therefore, it is reasonable to assume that any demand imbalance we see in the D2D market is associated with systemic issues in the D2C market that are not immediately apparent. Stocking effects may occur when market makers are risk-averse and these mismatches arise. (Menkhoff et al., 2025)

C. Conditional Order Imbalance and Foreign Exchange Volume (FXV)

In this portion, we speak about what the inventory model suggests will happen. It appears that the premium for speed is larger when there are more order gaps and a lot higher when risk-averse market makers seek a bigger premium for taking on more pricing risk when there is more ambiguity.

To achieve this goal, we double sort the daily reversal returns specified in the Europe window into six portfolios according to prefix net DD (low, medium, and high) and FXV (low and high). We employ the same idea of order flow as previously, suggesting that positive order flow reflects low net DD and negative order flow implies high net DD. The FXV is computed as the sum of squared returns of the previous day at the one-minute sampling frequency to offer an hourly total. The reverse returns, average anomalies, and volatilities for the GBP and AUD stocks are shown in Table VIII.

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608 **Table VIII.** Double Sorts: Dollar Demand and FX Volatility

609 This table is ordered by dollar demand (DD) at the European prefix hours; the sorting was done
610 by foreign exchange volatility (FXV) the day before. Discover the outcome of the plans to counter
611 the European fixings on the Australian dollar and the British pound. The daily reversal findings
612 are first categorised as low, medium, or high. In the early hours of the European trading day, these
613 categories are arranged according to the demand for dollars. The amount of FXV in each set
614 determines whether the portfolio is considered high or low. Each state's average yearly FXV and
615 dollar demand are displayed in the first four columns. The first four columns display the average
616 yearly FXV and dollar demand for each state. in the first four columns. BPS displays the typical
617 profit from reversals. The prefix window contains orders with a value in millions of dollars. Money
618 is what the market wants. By squaring the one-minute returns from the previous day, FXV may be
619 calculated and shown as a percentage point over a year. The "H-L" row displays the difference
620 between the high and low portfolio returns. When the values are significantly different (at least at
621 the 5% confidence level), an asterisk (*) is used to indicate this. To find out what "reversing
622 techniques" are, see Note 2. Data from 2006–2019 is shown in the sample. For this data, we turned
623 to Refinitiv FX Matching and Refinitiv Tick History.

624 **Panel A:**

Panel A: AUD	Avg. DD	Avg. DD	Avg. FXV	Avg. FXV	Avg. Returns	Avg. Returns	H-L
	FXVL	FXVH	FXVL	FXVH	FXVL	FXVH	
DDL	217.19	265.97	6.51	28.56	−14.60	−26.13	−10.80*
DDM	−11.61	−8.73	5.38	20.87	0.62	2.69	2.04
DDH	−235.35	−291.57	6.14	20.16	15.99	31.46	14.67*
H-L					30.59*	57.59*	

625 **Panel B**

Panel B: GBP	Avg. DD	Avg. DD	Avg. FXV	Avg. FXV	Avg. Returns	Avg. Returns	H-L
	FXVL	FXVH	FXVL	FXVH	FXVL	FXVH	
DDL	440.79	438.68	3.32	13.97	−11.89	−25.14	−12.65*
DDM	−15.62	−13.70	3.42	11.38	2.58	7.14	4.43
DDH	−423.64	−438.93	3.54	11.36	22.12	30.87	7.65*
H-L					34.01*	56.02*	

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One observation is that, depending on the level of volatility, the net DD distribution appears to be rather symmetrical. This indicates that a starting DD is unlikely to occur in a more FXV condition or vice versa. Consistent with the unconditional prefix DD, the DDM buckets likewise exhibit negative order imbalances for the AUD and the GBP. The average values in Table VII are derived from the sum of the order gaps for all the grouped stocks.

Then, it becomes clear that increasing the amount of reversal returns is a constant result of increasing the FXV from low to high. These alterations are very statistically significant at certain DD levels. Thirdly, when we shift from prefix dollar supply (DDL) to prefix dollar demand (DDH) scenarios, we see that reversal returns become very positive. Furthermore, the likelihood of these changes being genuine is high. Assuming there is no imbalance (such as an excess of demand or supply of US dollars), inventory models predict that providing liquidity will often provide a positive return. One strategy we outline in the study is the return reversal, which entails purchasing US dollars prior to the adjustment and selling them afterwards. On the DDL buckets, the reversal returns are negative. For this reason, any approach based on the unconditional imbalance we discussed previously would always include the latter.

Potentially massive return reversals are possible given the market's volatility. Reversing a DDL for the GBP takes -12 to 25 bps, and for the AUD, it takes -15 to 25 bps. Regarding DDH, the Australian dollar may have them between 16 and 30 basis points, while the British pound can have them between 22 and 29 basis points off. For both low and high net dollar demand buckets, the return gains from days with low volatility to days with high volatility range from 33 per cent to 101 per cent. Both forms of instability result in much slower rates of DDM bucket tipping over. This finding is not surprising, especially since the gaps are small, particularly on days of low volatility; this observation aligns with an inventory management concept.

As predicted by models of Grossman and Miller, return reversals are larger in size when order mismatches are larger, and they are significantly larger when there is a great deal of uncertainty as assessed by FXV. (Bredin & Fountas, 2025)

D. Asymmetric Information versus Inventory Effects

Here we shall discuss a regression definition associated with the work of Campbell, Grossman, Wang, Andrade, Chang, and Seasholes, as well as their price pressure forecasts. Their models

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demonstrate a negative inventory impact linking the return to delayed order imbalances and a positive information effect linking the return to simultaneous order imbalances. "We suggest that, considering the combined effects of inventory risk and uneven knowledge, (Orlov, 2024)

$$\Delta s_t^{\text{post-L}} = \alpha + \beta_1 \text{OF}_t^{\text{pre}} + \beta_2 \text{OF}_t^{\text{post-L}} + \varepsilon_t, \quad (3)$$

where $\Delta s_t^{\text{post-l}}$ represents the return computed using the post-fix window on day t and OF $_t^{\text{post-L}}$ how the related orders are processed. The sequence of events preceding the fix of $_t^{\text{pre}}$ is evaluated via (i) the combination of the "pre-E" and "E-L" windows (referred to as "pre-L"), or (ii) the "pre-E" window alone.

Here is the data for your review: To start, you can see that every single one of the table's contemporaneous order flow coefficients is enormous and very positive. According to this theory, details on the methodology used to determine prices This theory includes details on the methodology used to determine prices in dealer order flow. in dealer order flow. Second, the lagged prefix order flow coefficient should always be large and negative, as predicted, and conditional reversals due to inventory changes around the adjustment are consistent with this prediction. It's intriguing that the correlation between prefix order flow and postfix returns remains economically and statistically significant when we exclude the three hours before the London adjustment and simply consider the order flow up to the ECB adjustment. Evidence from prior studies (e.g., Evans) shows that trading in the last minutes before the London fix is not the only cause of the stocking effects.

Table IX. Return Reversal Regressions around European Fixes

The outcomes of the regression model are detailed in this table. $\Delta s_t^{\text{post-L}} = \alpha + \beta_1 \text{OF}_t^{\text{pre}} + \beta_2$ $\text{OF}_t^{\text{post-L}} + \varepsilon_t$ represents the logarithm of spot returns for the Australian dollar and the British pound on day t, using volume-weighted average prices (VWAPs) during the postfix hours, OF $_t^{\text{pre-E}}$ and OF $_t^{\text{pre-L}}$ calculate the volume of orders placed during the prefix hours of European fix times, and $_t^{\text{post-L}}$ Does postfix affect order flow? Returns have their origins. Order flow, expressed in billions of dollars, is the amount of orders placed by buyers minus the number of orders placed by sellers. Note 1 provides the meaning of the intraday periods. In brackets you can see the common errors

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that Newey and West corrected. The research period begins in June 2006 and ends in December 2019. Refinitiv FX Matching is the source of the data.

	AUD		GBP	
α	1.22	1.08	0.91	0.91
	(2.37)	(2.09)	(2.13)	(2.15)
OF $^{pre\text{-}E}$	−7.22		−3.54	
	(−2.79)		(−3.78)	
OF $^{pre\text{-}L}$	−9.28		−4.16	−4.37
	(−5.88)		(−5.67)	(−5.95)
OF $^{post\text{-}L}$	106.26	114.19	46.86	50.04
	(24.40)	(26.20)	(18.09)	(19.35)
Adj-R^2	0.30	0.32	0.21	0.23
Obs.	3,406	3,576	3,406	3,576

A reversal of around 2.5 basis points occurs when an order imbalance of one standard deviation, or roughly 5.5% of the trading volume during that window, generates a lag of 4.2 in the delayed order flow before the London benchmark for the GBP. We report an average unconditional impact in the section, and this is greater than that. In the case of the Australian dollar, the findings are somewhat more startling: a shock to delayed order flow of one standard deviation leads to a reversal of more than 3.3 basis points, or 5.5% of trading activity during that period. (Mueller et al., 2024)

E. Liquidity Provision

To determine whether dealers are genuinely providing liquidity at the fixings, this section examines returns to liquidity provision.

We begin with a dealer who attempts to profit from the daytime predictability of the European fixes. They bet on the D2C market that the dollar would rise in value. $t = 1$ before selling their USD at the fix $t = 2$ Shorting the US dollar and ending the position at allows them to profit from both the postfix depreciation and the reversal of the deal. $t = 3$ In the D2C market, the trader is always the one who wants to sell foreign currency at the bid price, which means they are always

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the liquidity demander. $t = 3$ Crucially, they purchase the foreign currency at the requested price during the fix, $s^{a}_{2,D2C}$ sell US dollars at the asking price, or the equivalent. The returns that follow are the two halves of the reverse returns that, after accounting for transaction costs, are defined in equation.

$$\Delta s^{\text{pre}}_{t,LD} = s^{b}_{1,D2C} - s^{a}_{2,D2C} \text{and} \Delta s^{\text{post}}_{t,LD} = -s^{a}_{2,D2C} + s^{b}_{3,D2C}, \quad (4)$$

in this case the combined return is determined by adding the two legs.

We also include a dealer that offers liquidity in the D2C market in addition to the trader, purchasing foreign currencies for US dollars at the fix on average to satisfy the unconditional demand for US dollars. However, liquidity providers purchase the foreign currency at the (lower) offer price, compared to liquidity demanders. And thus for the moment let's assume the dealer buys USD first thing in the morning from the D2D market. $t = 1$ To align with the trader's hypothetical positions, let's assume that the dealer shorts the US dollar at the fix and closes out their D2D position in the postfix trading session. $t = 3$ Keep in mind that the liquidity provider sells foreign currency at the bid price while acting as the "aggressor" while sourcing US dollars in the D2D market. $s^{b}_{t,D2D}$ Only in the case of providing liquidity at the fix does the dealer buy the foreign currency at the bid (or, conversely, sell US dollars at the ask). The ultimate returns to the two legs and the liquidity provider may be calculated as.

$$\Delta s^{\text{pre}}_{t,LP} = s^{b}_{1,D2D} - s^{b}_{2,D2C} \text{and} \Delta s^{\text{post}}_{t,LP} = -s^{b}_{2,D2C} + s^{b}_{3,D2D}, \quad (5)$$

Again again the total return is determined as the sum of the two legs. The table displays the results for both a liquidity provider and a liquidity demander, as stated in the equations, without relying on any external factors. The table displays the results for a liquidity provider and a liquidity demander, as stated in the equations, without relying on any external factors. The table presents the results for a liquidity provider and a liquidity demander, as stated in the equations, without relying on any external factors. Below is a summary of the findings. The turnaround becomes negative for the seller or liquidity demander when transaction fees are included first. You can't just arbitrage the unconditional return patterns if you're a trader looking to profit on intraday reversals. The second and foremost advantage of liquidity provision is that it is consistent with the concept that dealers provide financial support in return for a premium on liquidity. It is reasonable to assume that dealers will use streaming quotations to set prices and that customers will choose US

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dollars before the fixes, given that both prefix returns for liquidity provision are substantially positive. Buyers would anticipate losing money on an irreversible contract proceeding in this manner if they really want to acquire US dollars promptly, rather than the other way around. (Mancini et al., 2024)

Table X. Trading Venues, Reversal Portfolios, and Transaction Costs

This table displays the potential profits for a liquidity provider (LP) prior to the adjustment if they were to purchase foreign currency at the current bid in the D2D market and sell it at the current bid in the D2C market. Transaction fees are not included in the returns. You may see a liquidity demander in column "LD." This trader begins the prefix period by purchasing foreign currency in the D2C market and ends the fix by selling it in the same market sector. The postfix period is characterized by the reversal of all transactions. Their foundation is a bi-directional examination of the average returns of liquidity providers and consumers. Following the L closing at 16:00, trading ensues. There is a lack of depth in Reuter's tools. Returns are shown in BPS units. Quotations indicate the t-statistics. Two, a discussion of reverse procedures is available here (Note 2). Data from 2006–2019 is shown in the sample. For this data, we turned to Refinitiv FX Matching and Refinitiv Tick History.

	AUD			GBP		
	LP	**LD**	***p*-Val**	**LP**	**LD**	***p*-Val**
$\Delta s^{\text{pre-E}}$	0.60	−4.51	0.00	1.89	−0.54	0.01
	(0.87)	(−6.59)		(2.86)	(−0.81)	
$\Delta s^{\text{pre-L}}$	1.23	−3.70	0.00	2.15	−0.28	0.04
	(1.35)	(−4.10)		(2.55)	(−0.34)	
$\Delta s^{\text{post-L}}$	1.85	−2.83	0.00	2.06	−0.44	0.00
	(2.68)	(−4.14)		(4.61)	(−0.98)	

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	AUD			GBP		
	LP	**LD**	***p*-Val**	**LP**	**LD**	***p*-Val**
$\Delta s^{\text{pre-E+post-L}}$	2.45	−7.34	0.00	3.95	−0.98	0.00
	(2.55)	(−7.72)		(4.83)	(−1.19)	
$\Delta s^{\text{pre-L+post-L}}$	3.08	−6.53	0.00	4.21	−0.73	0.00
	(2.74)	(−5.86)		(4.32)	(−0.74)	

F. Discussion of Empirical Findings

Our evidence is consistent with an inventory risk hypothesis and is based on data from both direct-to-consumer and direct-to-device systems. We prove that (i) at the fix, demand for dollars is unconditional, (ii) returns are more volatile in the presence of larger order imbalances and more uncertainty, and (iii) projected returns to liquidity provision are large. According to these findings, dealers aren't the ones who demand US dollars from their customers, but rather the ones who lend them to those in need.

Our study refrains from discussing the origin of this unconditional demand, even though it is consistent with an unconditional V-shaped reversal pattern that the unconditional imbalance fits. According to our research (e.g., Jiang, Krishnamurthy, and Lustig), the constant demand for US dollars at the benchmark could be explained by the fact that the US is the global supplier of safe assets and the dollar is the reserve currency. Furthermore, the data we know on intraday stock market reversals aligns with the anticipated demand for speed at certain periods of the day (see Heston, Korajczyk, and Sadka as well as Lou, Polk, and Skouras). (Jiang et al., 2024; Hassan et al., 2024)

VII. Conclusion

In this paper, we discover a new model fact about how returns change during the day on currency markets: the US dollar slowly goes up before the three big currency fixes and then goes down

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afterward. Thus Tokyo, ECB, and London fixings capped the value of the US dollar during the day.

We use a set of natural experiments to look at how important published benchmark rates are for intraday reversal patterns. We also show that reversals around fixes are a regular trait of the data. They can be seen for all G9 currencies, which constitute about 75% of the world's spot trade volume over the 21 years that our data spans. Both the exchange-traded futures market and the over-the-counter forward market The trends in the results influence both the exchange-traded futures market and the over-the-counter forward market. by the trends in the results. This finding is interesting for big investors and corporate hedgers because it means that the time of their trading and speculation activities during the day, as well as currency changes and portfolio adjustments, all have an impact on their balance sheets. (Ahmad et al., 2024; Baele et al., 2025)

We don't agree with a few other ideas and present evidence that supports an account based on the need for U.S. cash at the time and the desire for immediate action. Based on volume data from a D2D market, we observe that there is an unbalanced order in favor of US dollars before the fixes. As order mismatch and unpredictability rise, FXV shows more and more reverse returns. Also, dealers who buy US dollars in the D2D market and act as liquidity providers at the fix may make a lot of money by doing so. This last point is that the numbers support an inventory management view, but we still need to find out why there is a trend toward the U.S. prefix DD. They are Wei Xiong, Amit Seru, Philip Bond, and Stefan Nagel., (Menkhoff et al., 2025; Hancu & Ranaldo, 2025; Ranaldo & Santucci de Magistris, 2025)

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